

4

Change of State

In this chapter, we will learn about

- latent heat and specific latent heat
- evaporation and condensation

4.1 Latent heat

Substances in daily life exist in three states: solid, liquid and gas (Fig 4.1a). In this chapter, we will study the energy change when a substance changes from one state to another.

Flipped classroom

Watch a video and answer the questions.

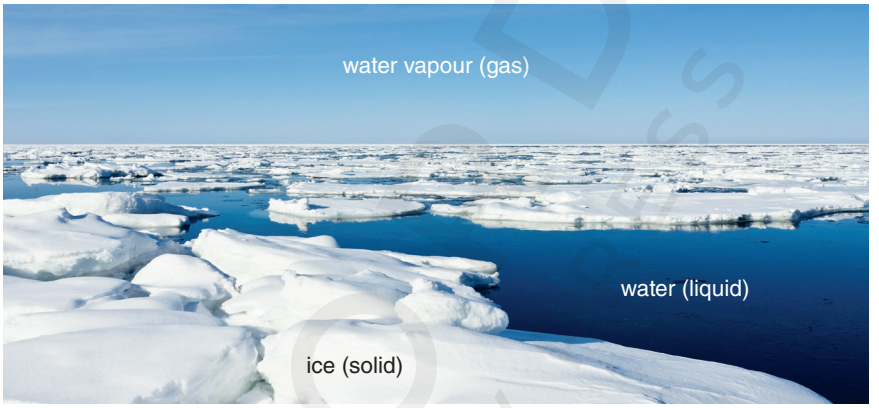


Fig 4.1a *Examples of gas, liquid and solid.*

DSE goal

State the three states of matter.

Determine the melting point and boiling point.

Realize latent heat as the energy transferred during the change of state without temperature change.

Video

→ This video demonstrates the Hands-on activity.

1 Latent heat and change of state

a Absorbing latent heat in fusion and boiling

How do the temperature change?

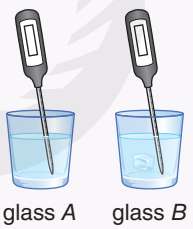
Hands-on activity

→ Worksheet is available on OUP web.

What you need



What to do



1. Put an ice cube into glass *B* and fill the glass with water. Stir the water and wait for a few minutes. Do not make the ice cube completely melt.
2. Fill glass *A* with water to about the same level as glass *B*.
3. Put the two thermometers into the two glasses respectively. Compare the temperatures in the two glasses every minute for the next 10 minutes.

What you find

How do the temperatures in the two glasses change?

In the activity, both the water and the iced water are cooler than their surroundings, so they keep absorbing energy. While the temperature of the water in glass *A* increases continuously from 0 °C, the iced water in glass *B* remains at 0 °C for a certain period of time and the ice continues to melt (Fig 4.1b).

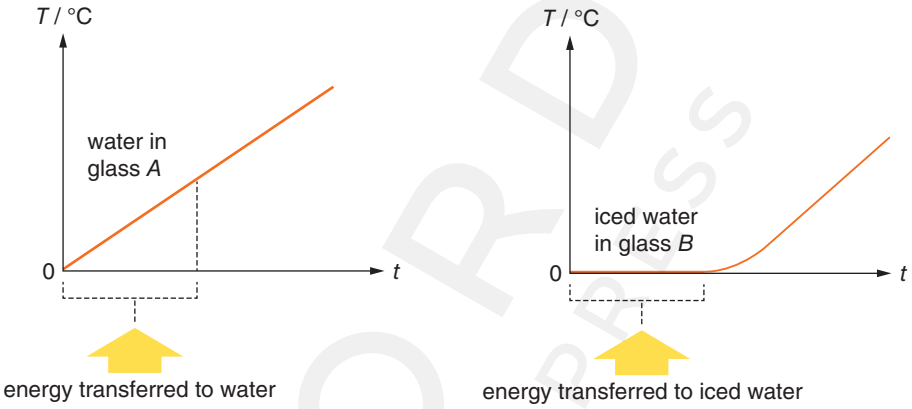


Fig 4.1b *Changes in temperature T in glasses A and B over time t*

In fact, a substance absorbs energy when it changes from a solid state to a liquid state, i.e. when **fusion** (or melting) occurs. Fusion occurs at the melting point of a substance. The temperature of the substance does not increase during fusion. The energy absorbed by a substance during a change of state is called **latent heat** (Fig 4.1c).

'Latent' means hidden. ▶

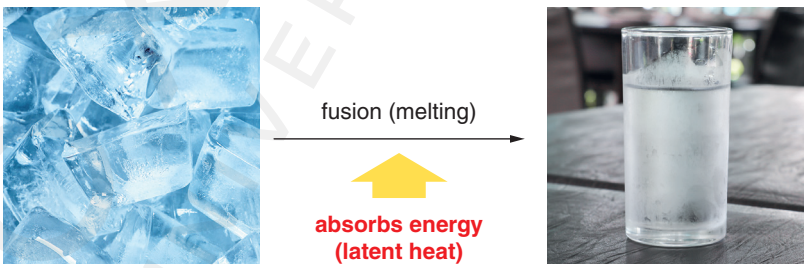


Fig 4.1c *A substance absorbs latent heat during fusion without a change in temperature.*

Boiling occurs when a substance changes from a liquid state to a gaseous state at the boiling point. During boiling, latent heat is absorbed without an increase in temperature (Fig 4.1d).

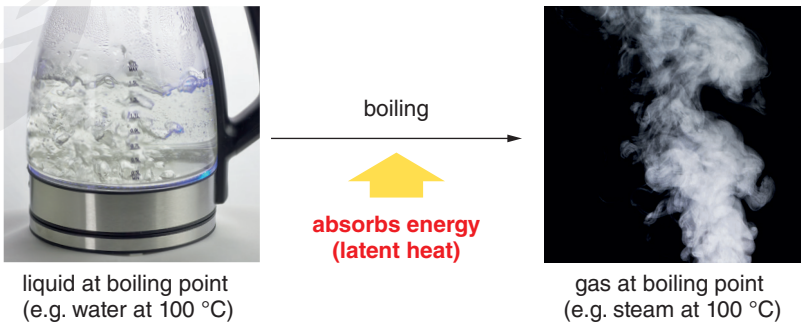
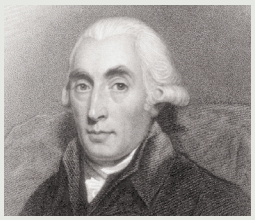


Fig 4.1d *A substance absorbs latent heat during boiling without a change in temperature.*

Steam is colourless and cannot be seen. The white 'fog' is observed in the photo because the steam changes back to water after it meets the colder surroundings. ▶

Know more

Joseph Black (1728–1799)



The idea of latent heat was introduced by *Joseph Black* in the 1760s. Black was a Scottish physicist and chemist. He also contributed significantly to the study of the specific heat capacity of a substance.

fusion 熔解　latent heat 潛熱　boiling 沸騰　Joseph Black 布拉克

In general, **vaporization** is the process in which a substance changes from a liquid state to a gaseous state. Latent heat is absorbed during vaporization. Boiling and evaporation are the two types of vaporization. Unlike boiling, evaporation can occur below the boiling point. You will learn more about evaporation in Chapter 4.2.

When a substance melts or vaporizes, it absorbs latent heat.

→ This video demonstrates Expt 4a.

Video

Let us do Experiment 4a to study how the temperature of a substance changes when it is heated continuously and undergoes changes in state.

Experiment 4a / Heating curve

The set-up is very hot during heating. Do not touch it.

1 Put some crushed ice and a temperature sensor into a beaker. Heat the beaker using a Bunsen burner.

2 Obtain a temperature–time graph to observe the temperature change of the ice/water during heating.



→ This is a Video Quiz about the experiment 'Heating curve of octadecan-1-ol'.

Discussion

Does the temperature keep increasing during heating? At what temperature does the ice turn into water and the water turn into steam?

From the above experiment, the following curve can be obtained (Fig 4.1e). This temperature–time graph is a **heating curve** of ice.

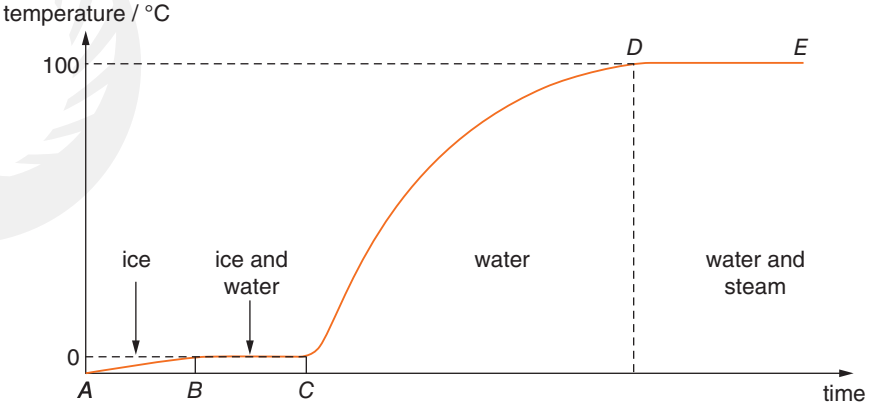


Fig 4.1e *A heating curve of ice.*

vaporization 汽化　　heating curve 加熱曲線

The heating curves of different substances have similar shapes.

1 *AB*: The temperature of the **solid** rises.

2 *BC*: The temperature remains constant at the **melting point**. In this stage, the solid is melting and is turning into liquid. The substance is a **solid-liquid mixture**. It is absorbing latent heat without a rise in temperature.

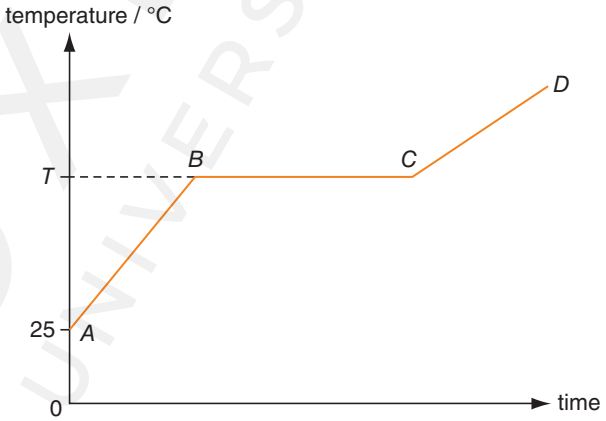
3 *CD*: All the solid has become liquid. The temperature of the **liquid** rises.

The gas (steam) will rise and leave the beaker.

4 *DE*: The temperature remains constant at the **boiling point**. In this stage, the liquid is boiling and is turning into gas. Both **liquid** and **gas** exist. Again, the substance is absorbing latent heat without a rise in temperature.

Example 1 Heating curve of water

Some water is heated at a constant power. The variation of the temperature of the water with time is as shown.



(a) Briefly explain why the graph is flat at stage *BC*.

(b) Does the water absorb energy at stage *BC*?

(c) Suggest a value for *T*.

Solution

(a) The water is boiling at stage *BC*. Therefore, its temperature remains constant.

(b) Yes

(c) 100 °C

Tip
T is the temperature when the water boils.

b Releasing latent heat in freezing and condensation

→ This video demonstrates Expt 4b.

Video

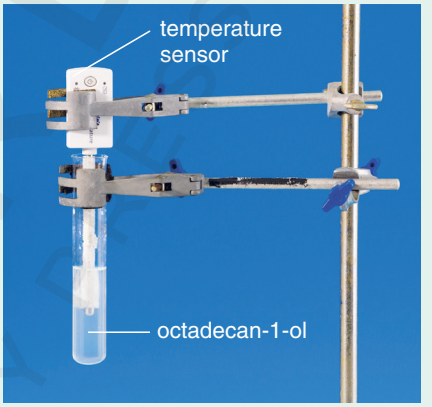
In Experiment 4b, we will study what happens when a substance is cooled.

Experiment 4b / Cooling curve

The set-up is hot. Do not touch it with bare hands.

1 Completely melt some octadecan-1-ol in a boiling tube using a water bath.

2 Remove the water bath. Measure the temperature of octadecan-1-ol with a temperature sensor. Obtain a temperature–time graph to observe the temperature change of octadecan-1-ol during cooling.



Discussion

Does the temperature keep decreasing throughout the cooling process?

Simulation

→ This simulation plots the cooling curve of water as it is cooled from steam to water and to ice. It also plots the heating curve from ice to water and to steam.

From the experiment, a **cooling curve** can be obtained (Fig 4.1f). Although the substance keeps releasing energy to the surroundings (Fig 4.1g), there is a period of time (*BC*) in which its temperature does not drop. During this period of time, the substance is changing from a liquid state to a solid state. This process is called **freezing** (or **solidification**), which occurs at the melting point (sometimes called freezing point). Both liquid and solid exist in this stage. Latent heat is released without a drop in temperature.

Every substance behaves similarly when they cool down from a liquid state.

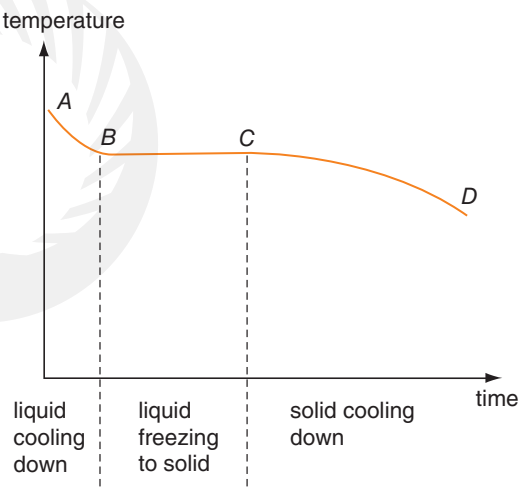


Fig 4.1f A cooling curve of octadecan-1-ol.

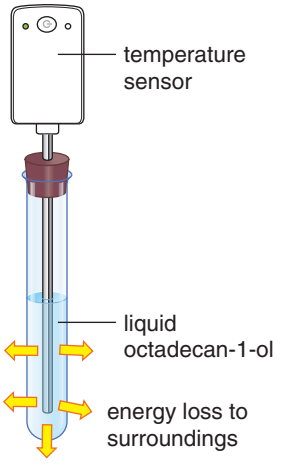


Fig 4.1g Energy is released by octadecan-1-ol continuously.

cooling curve 冷卻曲線　　freezing/solidification 凝固

Condensation is the process in which a substance changes from a gaseous state to a liquid state. Latent heat is also released during condensation.

Condensation occurs when a hot substance is cooled to its boiling point. During the process, most of the substance changes from a gaseous state to a liquid state without a drop in temperature.

Condensation can also occur below the boiling point of a substance. You will learn more about this in Chapter 4.2.

When a substance freezes or condenses, it releases latent heat.

Figure 4.1h summarizes the absorption and the release of latent heat during a change of state.

The energy absorbed or released during the conversion between the solid and liquid states is called the **latent heat of fusion**. The energy absorbed or released during the conversion between the liquid and gaseous states is called the **latent heat of vaporization**.

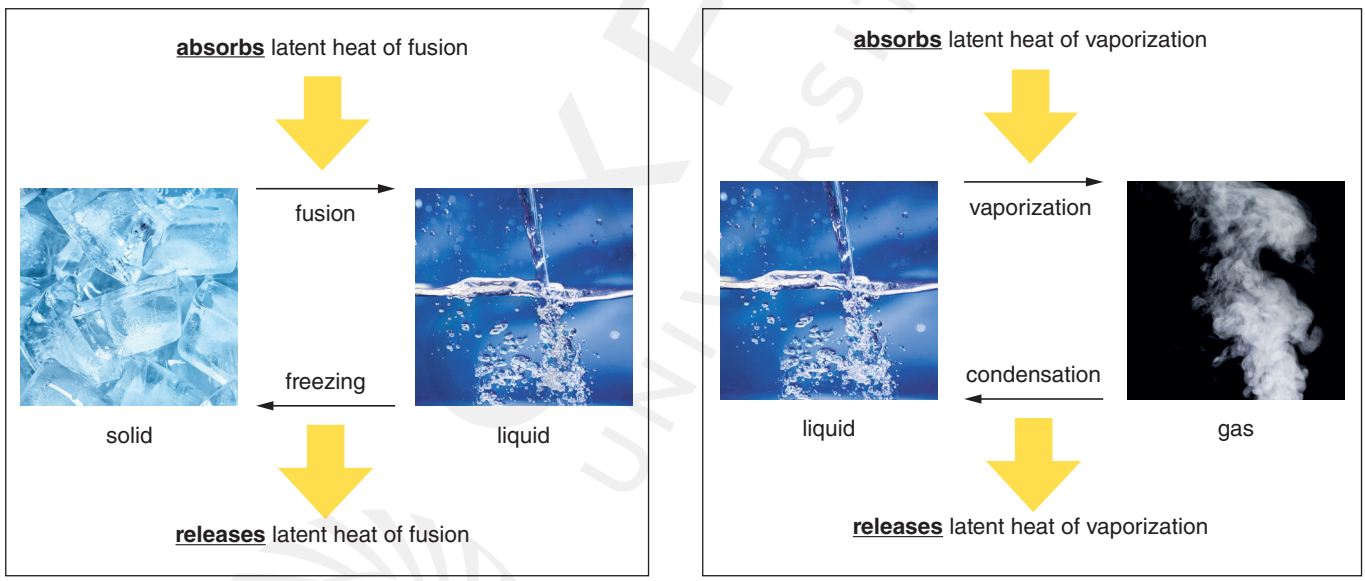


Fig 4.1h *Absorption and release of latent heat during a change of state.*

Know more

How does a rice cooker work?

First, the rice cooker is filled with uncooked rice and water. Energy is supplied to raise the temperature of the water to 100 °C. The temperature then remains at 100 °C as the water inside changes to steam and is absorbed by the rice. When all the water has been boiled away, the temperature rises again. As the electronic devices inside the rice cooker detect the temperature rise, the cooker stops cooking.



condensation 凝結

Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 Complete the table below.

	Process	Start state	End state	Is energy absorbed or released in the process?
(a)	Freezing			
(b)	Vaporization			
(c)	Fusion			
(d)	Condensation			

2 Refer to the following heating curve of octadecan-1-ol (Fig a) and complete Table a.

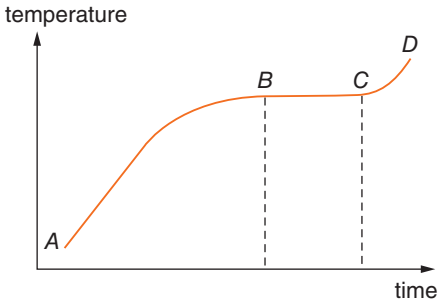


Fig a

Stage	*AB*	*BC*	*CD*
What is the state of octadecan-1-ol?			Liquid
How does the temperature change?	Increases		
Is energy absorbed?			Yes

Table a

DSE goal

Define specific latent heat of fusion as $l_f = \frac{Q}{m}$.

Define specific latent heat of vaporization as $l_v = \frac{Q}{m}$.

The **latent heat** is the energy needed to change the state of a certain mass of a substance. Its value depends on mass.

On the other hand, the **specific latent heat** refers to the energy needed to change the state of each kg of a substance. Its value does not depend on mass.

When explaining the above difference, Ts may refer to the discussion of the difference between heat capacity of an object and specific heat capacity of a substance in Ch 3.

2 Specific latent heat

The latent heat is the total energy absorbed or released by a substance during a change of state and is directly proportional to its mass. To compare whether a substance needs more energy to change its state than another substance, we should consider the **specific latent heat** of a substance. Specific latent heat is denoted by l, its unit is J kg^{-1}.

The specific latent heat of a substance is the energy absorbed or released to change the state of 1 kg of the substance without a change in temperature.

specific latent heat of substance (J kg^{-1})

latent heat needed for substance to change state (J)

$$l = \frac{Q}{m}$$

mass of substance (kg)

or

$$Q = ml$$

specific latent heat 比潛熱

The **specific latent heat of fusion** l_f of ice is the energy needed to change 1 kg of ice to water without a change in temperature. Its value is about 3.34×10^5 J kg^{-1}.

The **specific latent heat of vaporization** l_v of water is the energy needed to change 1 kg of water to steam without a change in temperature. Its value is about 2.26×10^6 J kg^{-1}.

Example 2 Energy needed to melt ice

Find the energy needed to melt 0.25 kg of ice at 0 °C. Given that the specific latent heat of fusion of ice is 3.34×10^5 J kg^{-1}.

Solution

Energy required = $ml_f = 0.25 \times (3.34 \times 10^5) = 8.35 \times 10^4$ J

The energy change of the ice in Example 2 can be illustrated by a simple diagram as shown in Figure 4.1i. The red arrow represents the energy gained by the ice. To distinguish from the energy gained causing a temperature change (which is represented by a vertical arrow), a horizontal arrow is used to represent the change of state.

This, together with the representation in Ch 3 p.68, will be used to represent the change in state and temperature in the later part of this book.

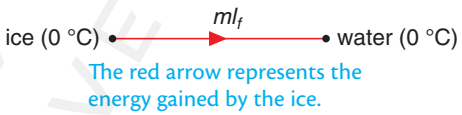
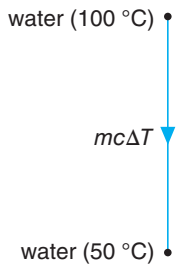


Fig 4.1i *Simple diagram illustrating the energy change of the ice.*

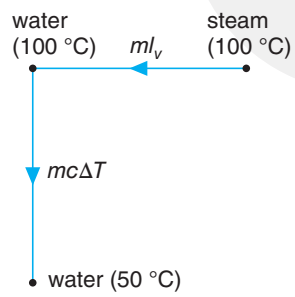
Tip

Simple diagram:



Tip

Simple diagram:



Example 3 Steam and hot water

Find the energy released when

(a) water of mass 15 g at 100 °C is cooled to 50 °C;

(b) steam of mass 15 g at 100 °C is cooled to water at 50 °C.

Given that the specific heat capacity of water is 4200 J kg^{-1} °C^{-1} and the specific latent heat of vaporization of water is 2.26×10^6 J kg^{-1}.

Solution

(a) Energy released = $mc\Delta T = 0.015 \times 4200 \times (100 - 50) = 3150$ J

(b) Energy released = $ml_v + mc\Delta T$

$= 0.015 \times (2.26 \times 10^6) + 3150$

$= 37\,050$ J

specific latent heat of fusion 熔解比潛熱 specific latent heat of vaporization 汽化比潛熱

Example 3 shows that the steam releases much more energy than the boiling water when both are cooled to 50 °C. Therefore, steam is more likely to cause serious burns than boiling water of the same mass.

Know more

Protecting plants from frost damage

In cold weather, plants may be damaged when the water in the plant cells freezes. To protect plants from frost damage, we can spray them with water.

Water has a high specific heat capacity and latent heat of fusion. Before water freezes, a large amount of latent heat is released to warm the plant. Also, the ice coating on the plant will insulate it from the cold surroundings.



Learn by practice 1

Take c (water) = 4200 J kg^{-1} °C^{-1}

l_f (ice) = 3.34×10^5 J kg^{-1}

l_v (water) = 2.26×10^6 J kg^{-1}

1 A heater transfers 1.36×10^5 J of energy to some water at 100 °C. Find the mass of water that is converted to steam.

2 0.7 kg of water at 10 °C is frozen to ice at 0 °C. How much energy is removed from it?

3 A beaker contains 60 g of ice at 0 °C (Fig a).

(a) How much energy is required to melt the ice?

(b) Find the total energy required to change the ice at 0 °C to water at 4 °C.



Fig a

4 Find the energy needed to change 1 kg of ice at 0 °C to steam at 100 °C (Fig b).

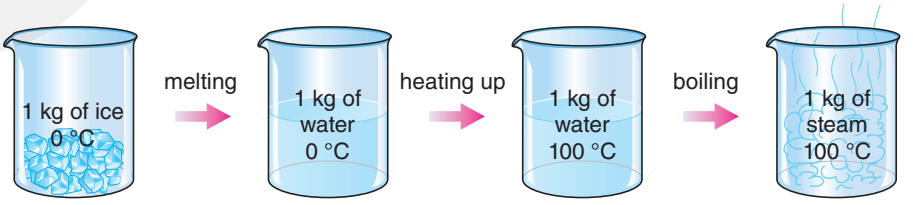


Fig b

3 Measuring specific latent heat

DSE exam

13(1A)Q2, 21(1B)Q1(a), 23(1A)Q2

Simulation Video

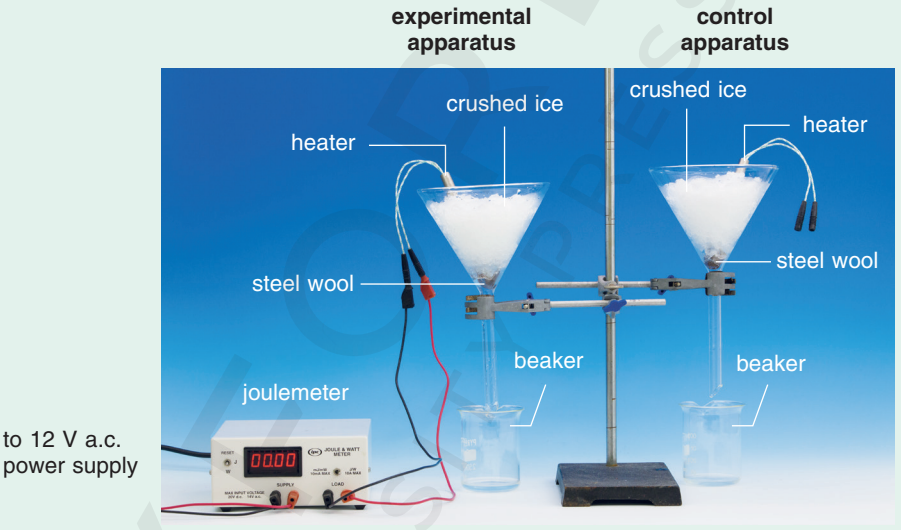
→ This simulation is a 'virtual expt' on the measurement of the specific latent heat of fusion of ice.

→ This video demonstrates Expt 4c.

Do not touch the heating part of the heater until it has cooled down.

Experiment 4c Measuring the specific latent heat of fusion of ice (I)

1 Set up the apparatus as shown. Fill both funnels with roughly equal amounts of crushed melting ice.



2 Switch on the heater for a period of time. Calculate the mass of ice melted by the heater. Read the joulemeter to measure the energy supplied to the heater.

3 Calculate the specific latent heat of fusion of ice from the results.

Precaution

1 Ice should be crushed to increase the contact area with the heater.

2 The heating parts of the heaters should be totally immersed in the ice.

3 Ensure melting ice is used so that it is at 0 °C and no energy is used to raise its temperature before melting.

4 Before switching on the heater, pack the crushed ice in the two funnels so that the drip rates are steady and about the same.

5 After switching off the heater, do not remove the beakers; wait until the drip rates are steady and about the same.

6 A small piece of wire gauze or steel wool in the neck of the funnels can prevent the crushed ice from dropping directly into the beakers.

Discussion

1 Compare the experiment result with the standard value 3.34×10^5 J kg^{-1}.

2 What are the possible sources of error in this experiment?

Some ice absorbs energy from the surroundings and melts (Fig 4.1j). By setting up the control apparatus, the effect due to any energy gained from the surroundings can be eliminated.

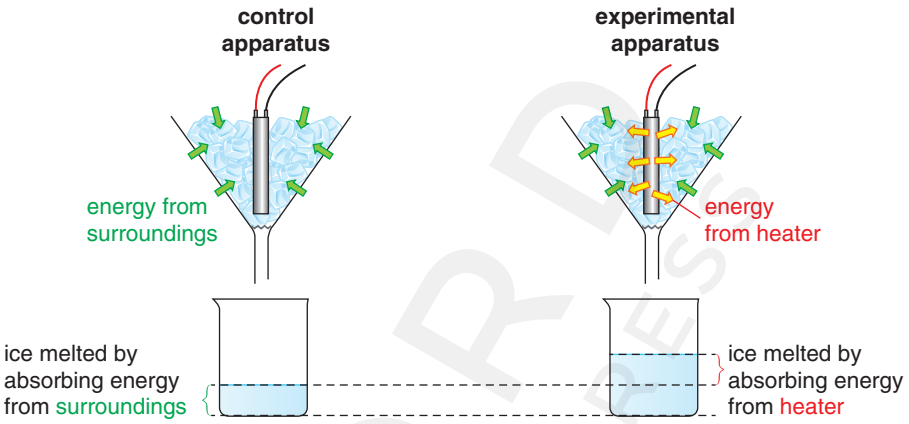


Fig 4.1j *The mass of ice melted by the heater can be found by setting up a control apparatus.*

Example 4 Finding the specific latent heat of fusion of ice

Jenny carries out Experiment 4c and obtains the following results:

Mass of water collected in experiment beaker $m_1 = 0.050$ kg

Mass of water collected in control beaker $m_2 = 0.014$ kg

Energy supplied by heater = 14 200 J

(a) Write the equation used to calculate the specific latent heat of fusion l_f of ice in the experiment.

(b) Calculate l_f using the above data.

(c) The standard value of l_f is 3.34×10^5 J kg^{-1}. Find the percentage error of the result in (b).

(d) State a possible source of error.

Solution

(a) $l_f = \dfrac{Q}{m}$

(b) Mass of ice melted by heater = $m_1 - m_2 = 0.050 - 0.014 = 0.036$ kg

$$l_f = \frac{Q}{m} = \frac{14\ 200}{0.036} = 3.94 \times 10^5 \text{ J kg}^{-1}$$

(c) Percentage error = $\dfrac{3.94 \times 10^5 - 3.34 \times 10^5}{3.34 \times 10^5} \times 100\% = 18.0\%$

(d) Not all the energy supplied by the heater is absorbed by the ice.

Writing practice 1

Question

How will the value of l_f, i.e. the specific latent heat of fusion of ice, found in Experiment 4c be affected if the control apparatus is not set up? Explain your answer. (3 marks)

Answer

Some ice absorbs energy from the surroundings and melts. 1A

If the control apparatus is not set up, the mass of water collected in the beaker is larger than the mass of ice melted due to the heater. 1A

By $l_f = \frac{Q}{m}$, the value of l_f found will be lower (than the actual value). 1A

Practise yourself

1 John mistakenly used ice of −5 °C when he carried out Experiment 4c. Explain how this mistake affects the value of l_f found. (3 marks)

2 May forgot to put a small wire gauze in the funnel of the experimental apparatus when she carried out Experiment 4c. Explain how this mistake affects the value of l_f found. (3 marks)

Simulation Video

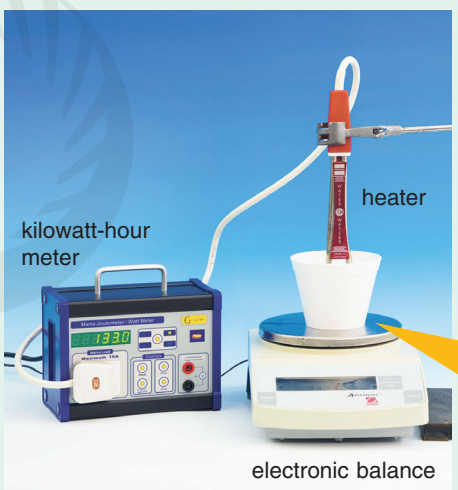
→ This simulation is a 'virtual expt' on the measurement of the specific latent heat of vaporization of water.

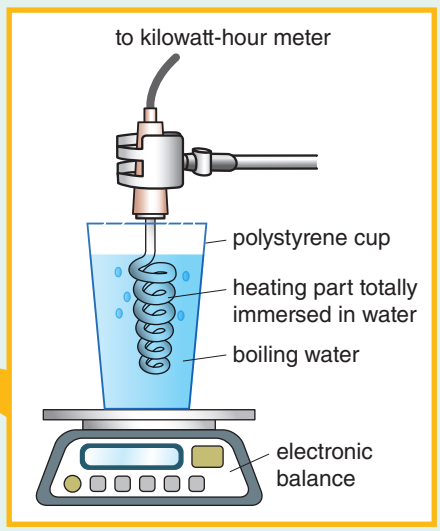
→ This video demonstrates Expt 4d.

- Do not touch the heating part of the heater until it has cooled down.
- Beware of the hot water.

Experiment 4d Measuring the specific latent heat of vaporization of water

1 Set up the apparatus as shown.

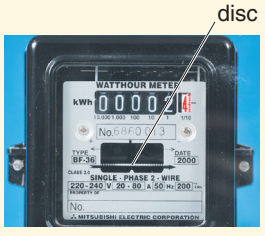




Skill
Experiment

Kilowatt-hour meter

The kilowatt-hour meter measures the energy supplied to the heater to boil the water. For the old type of kilowatt-hour meter, a disc on it rotates when it is in use. The energy used for each rotation is marked on the meter panel.
To measure the energy needed to boil the water, count the number of rotations of the disc.



An old type of kilowatt-hour meter.

→ Teaching notes and exercises are available on OUP web.

2 Switch on the heater. When the water boils, take the initial reading of the electronic balance. Zero the kilowatt-hour meter reading.

3 Turn off the heater after 0.025 kW h is used. Wait until the water becomes steady and take the final reading of the electronic balance.

4 Calculate the mass of water boiled away from the difference in the readings of the electronic balance.

5 Calculate the specific latent heat of vaporization of water from the results.

Precaution

Immerse the heating part of the heater totally in water before switching it on. Otherwise, the heater may overheat.

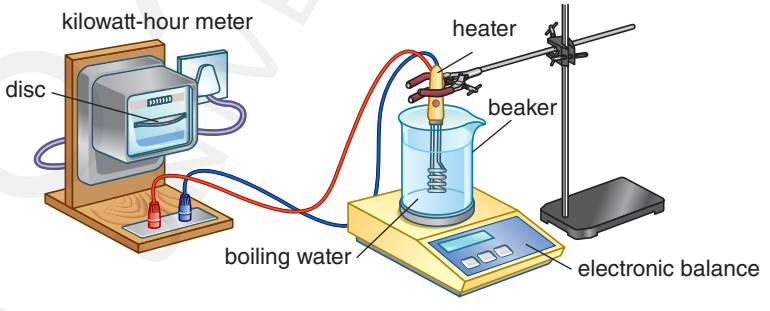
Discussion

1 Compare the experiment result with the standard value 2.26×10^6 J kg^{-1}.

2 What are the possible sources of error in this experiment?

Example 5 Finding the specific latent heat of vaporization of water

Jimmy carries out Experiment 4d using an old type of kilowatt-hour meter. With this type of kilowatt-hour meter, the energy supplied through it can be calculated from the number of rotations of the disc on it.



Energy supplied per revolution
$= \frac{1000 \text{ W} \times 3600 \text{ s}}{600}$
$= 6000$ J

The calibration of the kilowatt-hour meter is 600 rotations/kW h, meaning that the **energy supplied per revolution of the disc is 6000 J**.

The following results are obtained in the experiment.

Initial mass of water $m_1 = 0.82$ kg

Final mass of water $m_2 = 0.72$ kg

Number of rotations counted = 41

(a) Write the equation used to calculate the specific latent heat of vaporization of water l_v in the experiment.

(b) Calculate l_v using the above data.

(c) Jimmy suggests adding a lid to reduce experimental error. Explain why this is not a good suggestion.

Solution

(a) $l_v = \frac{Q}{m}$

(b) Mass of water vaporized = $m_1 - m_2 = 0.82 - 0.72 = 0.10$ kg

Energy supplied to vaporize the water = $6000 \times 41 = 246\ 000$ J

$l_v = \frac{Q}{m} = \frac{246\ 000}{0.10} = 2.46 \times 10^6$ J kg^{-1}

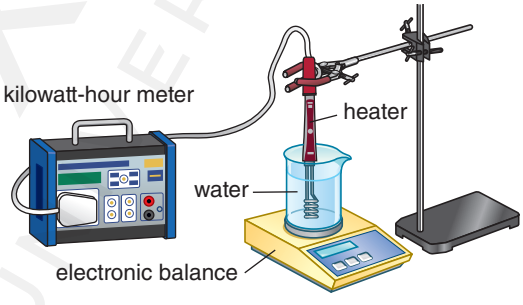
(c) Water vapour would condense back into water on the inner surface of the lid and drip back into the beaker.

Writing practice 2

Question

In Experiment 4d, energy in the hot water is lost to the surroundings. Explain how this error affects the value of l_v found in the experiment.

(2 marks)



Answer

As some energy is lost to the surroundings, the mass of water boiled away is smaller than it should be. 1A

By $l_v = \frac{Q}{m}$, the value of l_v found will be higher. 1A

Practise yourself

1 In Experiment 4d, some water splashes out of the cup. Explain how this error affects the value of l_v found in the experiment. (2 marks)

2 In Experiment 4d, water at 25 °C is prepared. The initial kilowatt-hour meter reading is recorded when the water starts heating. Explain how this mistake affects the value of l_v found in the experiment. (3 marks)

Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 Match the following experiments (Fig a–c) with the precautions that can be taken to improve their accuracy by putting ticks in the table below.



Fig a *Finding l_v.*



Fig b *Finding l_f.*

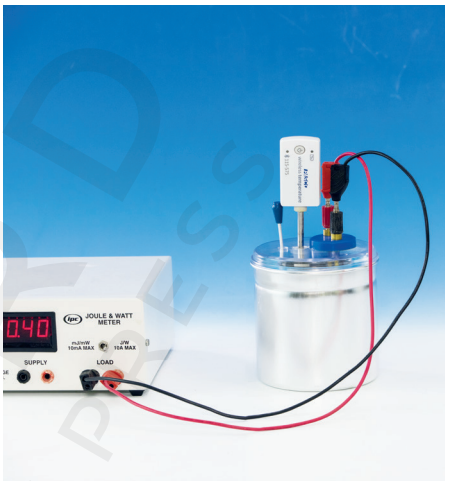


Fig c *Finding c.*

	Precaution	Finding l_v	Finding l_f	Finding c
(a)	Taking the temperature immediately after switching off the power supply			
(b)	Covering the container with a lid			
(c)	Stirring the sample throughout the experiment			
(d)	Using crushed ice			
(e)	Using melting ice			
(f)	Immersing the heating part of the heater into the ice/water completely			
(g)	Wrapping the container in cotton wool			

DSE goal

Solve problems involving latent heat.

DSE exam

12(1B)Q1, 14(1A)Q2, 16(1A)Q2, 17(1A)Q2, 19(1B)Q1(a), 21(1A)Q1, 23(1B)Q1

Unless otherwise specified, we assume that no energy is lost to the surroundings. ▶

4 Law of conservation of energy

In Chapter 3, we studied what happens when two bodies are put in contact. The two bodies will reach **thermal equilibrium**, i.e. the **same temperature**, and the energy lost by the hotter body is equal to the energy gained by the colder body. This is a result of the law of conservation of energy. After learning how the energy in a substance changes during a change of state, we are now able to solve problems of mixing substances involving changes of state.

Read more about steaming and simmering (another cooking method) on p.136. ▶

Example 6 Reheating the soup

Mary reheats 250 g of tomato soup at 4 °C using a steamer. In this process, 20 g of steam at 100 °C condenses on the tomato soup. The specific heat capacities of the tomato soup and water are 3700 J kg^{-1} °C^{-1} and 4200 J kg^{-1} °C^{-1} respectively. The specific latent heat of vaporization of water is 2.26×10^6 J kg^{-1}.



(a) Find the final temperature T of the tomato soup. Assume there is no energy loss to the surroundings.

(b) Suppose the energy loss to the surroundings is not negligible. Compare the actual final temperature with the result in (a).

Solution

(a) Energy lost by steam = energy gained by soup

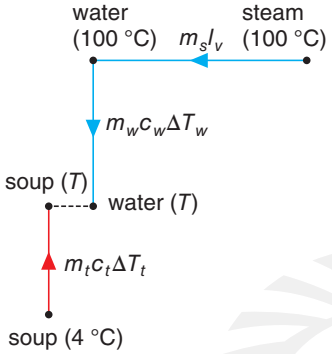
$$m_s l_v + m_w c_w \Delta T_w = m_t c_t \Delta T_t$$

$$0.02 \times (2.26 \times 10^6) + 0.02 \times 4200 \times (100 - T) = 0.25 \times 3700 \times (T - 4)$$

$$T = 56.8 \text{ °C}$$

(b) As some energy is lost to the surroundings, less energy would be absorbed by the tomato soup. Therefore, the final temperature would be lower.

Letters *s*, *w* and *t* printed in subscript represent steam, water and tomato soup respectively. ▶



Learn by practice 2

Take c (water) = 4200 J kg^{-1} °C^{-1}

l_f (ice) = 3.34×10^5 J kg^{-1}

l_v (water) = 2.26×10^6 J kg^{-1}

1 Simon adds 80 g of ice at 0 °C to 0.3 kg of soft drink at 25 °C. Find the final temperature of the mixture, given that the specific heat capacity of the soft drink is 5300 J kg^{-1} °C^{-1}.

2 Steam of mass 0.1 kg at 100 °C is mixed with 2 kg of water at 20 °C. What is the final temperature of the mixture?

3 Kelly prepares a cup of juice with a mass of 0.3 kg and a temperature of 68 °C. She then adds 100 g of ice at 0 °C to lower the temperature of the juice to 30 °C. Estimate the specific heat capacity of the juice.

By applying the law of conservation of energy, we can measure the specific latent heat of fusion of ice using a method different from that in Experiment 4c. This is shown in the following experiment.

Video

→ This video demonstrates Expt 4e.

Experiment 4e / Measuring the specific latent heat of fusion of ice (II)

1 Pour 200 g of water at 40 °C in a polystyrene cup.

2 Put a melting ice cube into the cup of water. Stir the water well.



3 After the ice melted, measure the lowest temperature reached by the water. Also measure the total mass of water to find the mass of the ice added.

4 Calculate the specific latent heat of fusion of ice from the results.

Discussion

1 Compare the experiment result with the standard value 3.34×10^5 J kg^{-1}.

2 What is the possible source of error in this experiment?

Sometimes, only part of a substance changes its state from one to another during mixing. We should notice when this happens.

☆ HKDSE 2016 Paper 1A Q2

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Mixing ice with juice

Amy adds 200 g of juice at 15 °C to a glass containing 50 g of ice at 0 °C. What is the final temperature of the drink? Assume the energy transferred to the glass and the surroundings is negligible. The specific heat capacities of the juice and water are 3730 J kg^{-1} °C^{-1} and 4200 J kg^{-1} °C^{-1} respectively. The specific latent heat of fusion of ice is 3.34×10^5 J kg^{-1}.



A –7.39 °C

B –5.76 °C

C 0 °C

D 5.76 °C

Solution

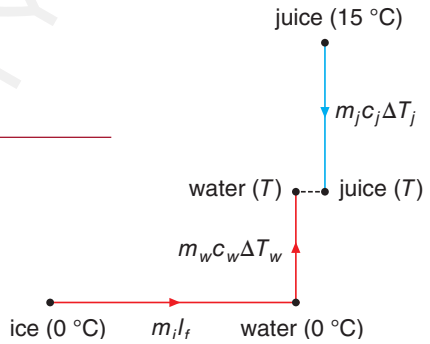
Assuming all the ice melts in the process, we will get:

energy lost by juice = energy gained by ice

$$m_j c_j \Delta T_j = m_i l_f + m_w c_w \Delta T_w$$

$$0.2 \times 3730 \times (15 - T) = 0.05 \times (3.34 \times 10^5) + 0.05 \times 4200 \times (T - 0)$$

$$T = -5.76 \text{ °C}$$



However, this is physically **not possible** as both the ice and the juice have initial temperatures higher than this value.

This suggests that the **assumption** that all the ice will melt into water and the temperature of the water will rise to T is **incorrect**. To check this:

Common mistake ⚠
Students may fail to realize that not all the ice melts when the mixture is in thermal equilibrium.

Energy needed to melt all the ice = $m_i l_f$

$$= 0.05 \times (3.34 \times 10^5)$$

$$= 16\,700 \text{ J}$$

Energy released to cool the juice to 0 °C = $m_j c_j \Delta T_j$

$$= 0.2 \times 3730 \times (15 - 0)$$

$$= 11\,190 \text{ J } (< 16\,700 \text{ J})$$

You may do this checking at the beginning to figure out what will happen before setting up an equation.

Therefore, when the temperature of the juice drops to 0 °C, not all the ice melts. This means that both the juice and the ice/water are at the same temperature of 0 °C and reach thermal equilibrium.

Two substances reach thermal equilibrium when they reach the same temperature. They do not need to be in the same state.

∴ The answer is C.

▶ Revision 4 Q10 (p.130)

The law of conservation of energy can also be applied to solve problems involving heating substances with a fixed power.

Example 7 Reheating vegetables

Mandy uses a steamer to reheat 800 g of vegetables at 25 °C.

Given:

specific latent heat of vaporization of water = 2.26×10^6 J kg^{-1}

specific heat capacity of water = 4200 J kg^{-1} °C^{-1}

specific heat capacity of the vegetables = 3000 J kg^{-1} °C^{-1}

When the water boils, Mandy puts the vegetables into the steamer for 10 minutes. Suppose half of the steam condenses on the vegetables, and energy is supplied to the water at 500 W.



(a) Find the total mass of steam condensing on the vegetables.

(b) Estimate the final temperature T of the vegetables.

Solution

(a) Energy supplied by steamer = energy gained by 100 °C water

$$Pt = ml_v$$

$$500 \times (10 \times 60) = m \times (2.26 \times 10^6)$$

$$m = 0.1062 \text{ kg}$$

$$\text{Total mass of steam condensing on the vegetables} = \frac{1}{2} \times 0.1062$$

$$= 0.0531 \text{ kg}$$

(b) Energy lost by steam = energy gained by vegetables

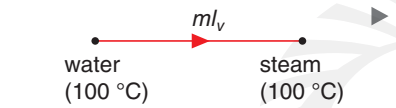
$$m_s l_v + m_w c_w \Delta T_w = m_v c_v \Delta T_v$$

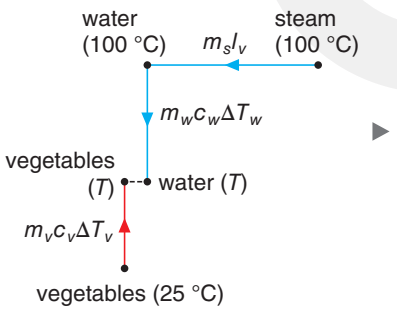
$$0.0531 \times (2.26 \times 10^6) + 0.0531 \times 4200 \times (100 - T) = 0.8 \times 3000 \times (T - 25)$$

$$T = 77.1 \text{ °C}$$

Tip

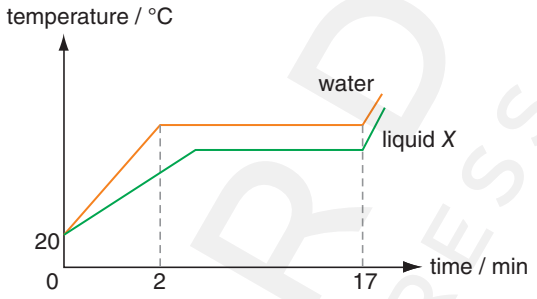
Water of mass m at 100 °C becomes steam at 100 °C in 10 minutes.





Example 8 Heating water

500 g of water is heated at a constant power P and a heating curve (orange line) is obtained as shown below.



Take the specific heat capacity of water as 4200 J kg^{-1} °C^{-1}.

(a) Find P.

(b) Hence, estimate the specific latent heat of vaporization of water l_v.

(c) Another liquid X of mass 500 g is heated at the same power. The heating curve (green line) is also shown in the figure. Compare the following quantities of liquid X with those of water and explain briefly.

(i) Boiling point

(ii) Specific heat capacity

(iii) Specific latent heat of vaporization

Solution

Tip
The temperature of water rises from 20 °C to 100 °C during 0–2 min.

(a) Energy supplied in heating = energy gained by water

$$Pt = mc\Delta T$$
$$P \times (2 \times 60) = 0.5 \times 4200 \times (100 - 20)$$
$$P = 1400 \text{ W}$$

Tip
All the water becomes steam during 2–17 min.

(b) Energy supplied in heating = energy gained by water

$$Pt = m_s l_v$$
$$1400 \times (15 \times 60) = 0.5 l_v$$
$$l_v = 2.52 \times 10^6 \text{ J kg}^{-1}$$

Tip
A higher specific heat capacity means more energy is needed to raise the temperature by 1 °C.

(c) (i) X boils at a lower temperature. Therefore, X has a lower boiling point.

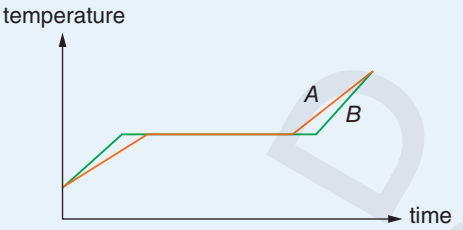
(ii) For the same mass heated at the same power, the temperature of X rises at a lower rate. Therefore, X has a higher specific heat capacity.

Tip
A higher specific latent heat means more energy is needed to change 1 kg of a substance from one state to another.

(iii) For the same mass heated at the same power, X takes less time to vaporize. Therefore, X has a lower specific latent heat of vaporization.

Right and wrong

Two substances *A* and *B* are melting under the same conditions.



✗ *A* must have a smaller specific latent heat of fusion than *B*.

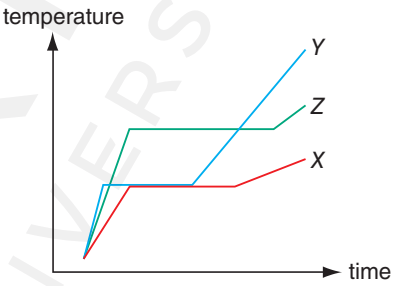
✓ It takes less time for *A* to melt. Therefore, *A* absorbs less latent heat than *B* during melting. However, the masses of *A* and *B* are not known, so their specific latent heat of fusion cannot be compared.

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

(For Q1–3.) Jimmy melts three materials *X*, *Y* and *Z* of equal mass by heating them at the same constant power. The figure below shows how the temperatures of *X*, *Y* and *Z* change with time.



1 Which material(s) has/have the highest melting point?

2 Which material(s) has/have the largest value of specific latent heat of fusion?

3 Which material(s) has/have the largest value of specific heat capacity in solid state?

Learn by practice 3

Take c (water) = 4200 J kg^{-1} °C^{-1}, l_v (water) = 2.26 × 10^6 J kg^{-1}.

1 A 1.5-kW heater is immersed in 2 kg of water at 90 °C. How long does it take to

(a) heat the water to 100 °C?

(b) vaporize 0.5 kg of the water?

2 A 1-kW heater, immersed in 0.5 kg of water at 20 °C, is switched on for 10 minutes. Calculate the amount of water that boils away.

DSE goal

Interpret latent heat in terms of the change of potential energy of the molecules during a change of state.

DSE exam

12(1A)Q4, 19(1A)Q2, 22(1A)Q2

Simulation

→ This simulation shows how the models of solid, liquid and gas change from one state to another as the temperature changes. It is available on OUP web in PC version (.exe) only.

5 Molecular potential energy and internal energy

We learned in Chapter 2 that molecules in motion have kinetic energy (KE). When the temperature of a body rises, its molecules move faster and the average KE of the molecules increases.

Also, there are attractive forces acting between the molecules. These forces give rise to the **potential energy** (PE) of the molecules. When a substance changes from a solid to a liquid or a liquid to a gas, its molecules absorb energy (latent heat) to overcome the attractive forces between them so that they have more freedom to move around (Fig 4.1k). In this process, the PE of the molecules increases but their average KE remains unchanged.

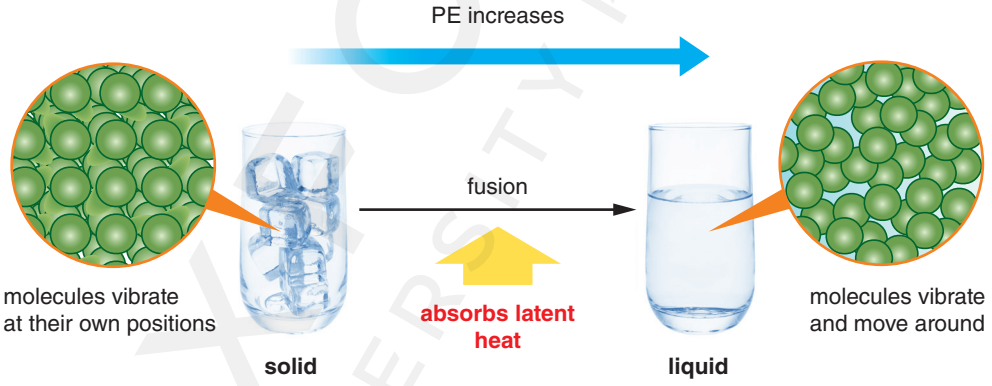


Fig 4.1k *Latent heat is needed to overcome the attractive forces among the molecules in a solid to break up their regular arrangement.*

Cross-link

Molecular KE is a part of the internal energy stored in a body. See Ch 2 p.23.

In fact, molecular PE is another part of the internal energy stored in a body. Therefore, after the body absorbs latent heat, its internal energy also increases.

The internal energy of a body is the sum of the kinetic energy and potential energy of all its molecules. It is the total energy stored in the body.

In other words,

$$\text{internal energy} = \text{total molecular KE} + \text{total molecular PE}$$
$$= \text{no. of molecules} \times (\text{avg. molecular KE} + \text{avg. molecular PE})$$

From this, we can see that internal energy is related to the mass, temperature and state of a body in the following way (Table 4.1a on p.114):

potential energy 勢能

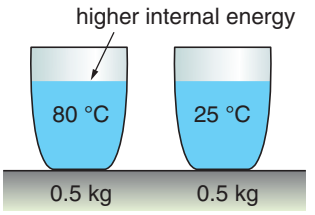
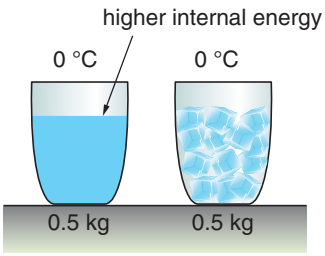
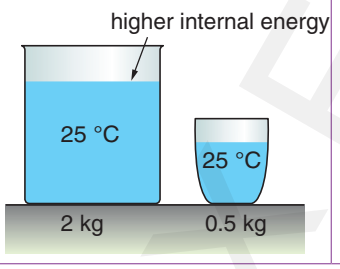
Factor	Average molecular KE	Average molecular PE	Internal energy
Temperature (average KE) 	hot water > cold water	the same	Hot water has a higher internal energy than cold water of the same mass.
State (average PE) 	the same	liquid > solid	Water at 0 °C has a higher internal energy than ice at 0 °C of the same mass.
Mass (no. of molecules) 	the same	the same	Water with a greater mass has a higher internal energy than water with a smaller mass at the same temperature.

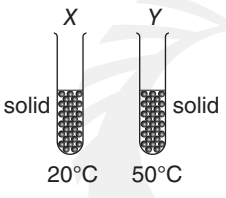
Table 4.1a *Factors affecting internal energy.*

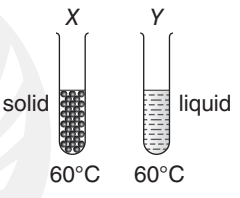
Google Forms Microsoft Forms

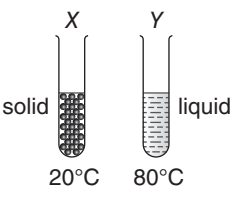
Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 In each of the following cases, test tubes *X* and *Y* hold the same mass of a material. The melting point of the material is 60 °C. Complete the table (by filling in '*X*', '*Y*', 'the same' or 'cannot be determined') to compare different kinds of energy in each case.

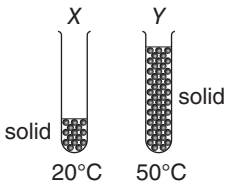
(a) 

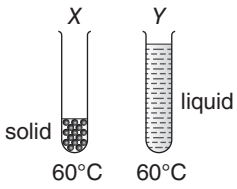
(b) 

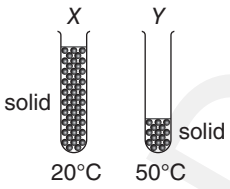
(c) 

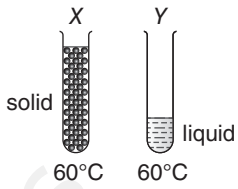
	In which test tube does the material have a higher				
	average molecular KE?	total molecular KE?	average molecular PE?	total molecular PE?	internal energy?
(a)					
(b)					
(c)					

2 In each of the following cases, test tubes *X* and *Y* hold different masses of a material. The melting point of the material is 60 °C. Complete the table (by filling in '*X*', '*Y*', 'the same' or 'cannot be determined') to compare different kinds of energy in each case.

(a)


(b)


(c)


(d)


	In which test tube does the material have a higher				
	average molecular KE?	**total molecular KE?**	**average molecular PE?**	**total molecular PE?**	**internal energy?**
(a)					
(b)					
(c)					
(d)					

Exercise 4.1

Take c (water) = 4200 J kg^{-1} °C^{-1}

l_f (ice) = 3.34×10^5 J kg^{-1}

l_v (water) = 2.26×10^6 J kg^{-1}

1 Which of the following processes releases latent heat?

A Boiling

B Melting

C Condensation

D Vaporization

2 Which of the following physical quantities is/are unchanged during the boiling process of a substance?

(1) Internal energy of the substance

(2) Average potential energy of the molecules

(3) Average kinetic energy of the molecules

A (3) only

B (1) and (2) only

C (1) and (3) only

D (1), (2) and (3)

★**3** *P* and *Q* are two solids made of different substances. They are heated at the same constant rate. Their heating curves are shown in Figure a.

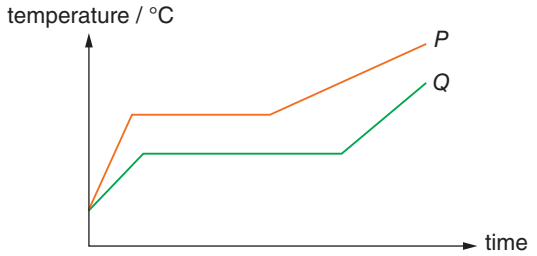


Fig a

Which of the following statements about *P* and *Q* must be correct?

(1) *P* absorbs less latent heat.

(2) *P* has a higher specific latent heat of fusion.

(3) *Q* has a lower melting point.

A (1) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

(For Q4–5.) John uses the apparatus below to find the specific latent heat of vaporization l_v of a liquid (Fig b).

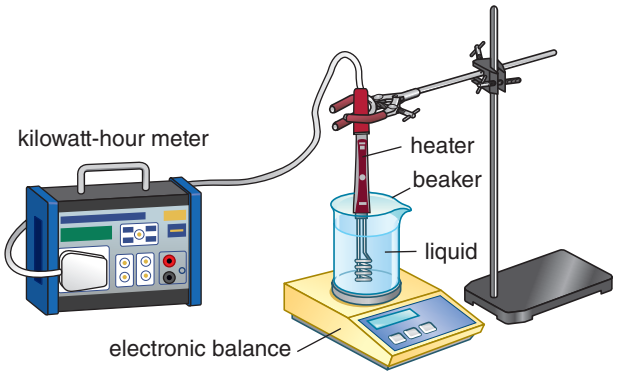


Fig b

★**4** Which of the following methods can improve the accuracy of his experiment?

(1) Cover the beaker with a lid.

(2) Wrap the beaker in cotton wool.

(3) Use a heater with a lower power.

A (1) only B (2) only

C (1) and (2) only D (1) and (3) only

★**5** Which of the following errors will cause the value of l_v obtained in the experiment to be larger than the standard value?

(1) Energy from the liquid is lost to the surroundings.

(2) Some liquid splashes out of the beaker.

(3) Some vapour condenses on the surface of the beaker and drips back into it.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★**6** Beakers P and Q contain the same material in solid state and liquid state respectively (Fig c). The mass in P is larger than the mass in Q.

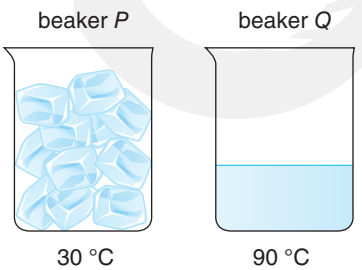


Fig c

Which of the following statements must be correct?

(1) The material in P has a lower internal energy.

(2) The material in P has a lower average molecular KE.

(3) The material in P has a lower average molecular PE.

A (1) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

7 A student wants to find the specific latent heat of fusion l_f of ice. He uses a 2-kW heater to heat some melting ice. 0.4 kg of water is obtained after heating for 1 minute.

(a) Estimate l_f from the above data.

(b) Suggest one precaution that can help improve the accuracy of the experiment.

8 Shirley pours 330 g of soft drink at 20 °C into a glass of ice at 0 °C. The mass of ice in the glass is 60 g. Find the final temperature of the mixture, given that the specific heat capacity of the soft drink is 3900 J kg^{-1} $°C^{-1}$. Assume that there is no energy exchange with the surroundings.

9 Water of mass m is heated from 0 °C to 100 °C. Then it starts to boil. The specific heat capacity of water is 4200 J kg^{-1} $°C^{-1}$. A student tries to find the specific latent heat of vaporization of water as shown (Fig d). Is the student's work correct? Explain your answer.

$\because Q = mc\Delta T$ and $Q = ml$

$\therefore l = c\Delta T$

$= 4200 \times 100$

$= 420\,000\ J\,kg^{-1}$

Fig d

★**10** Connie wants to prepare 1 kg of water at 100 °C by injecting steam at 100 °C into water at 25 °C.

(a) How much steam and water (in kg) does she need?

(b) State one assumption in your calculation.

(c) Would the actual mass of steam used be larger or smaller? Explain your answer.

*11 Which of the following would cool 0.3 kg of water at 50 °C to a lower temperature, a water bath of mass 1 kg at 0 °C or ice of mass 0.2 kg at 0 °C? Show your calculations.

*12 Tom heats 1.2 kg of a solid of a certain substance at a constant power of 1200 W and obtains a heating curve as shown (Fig e).

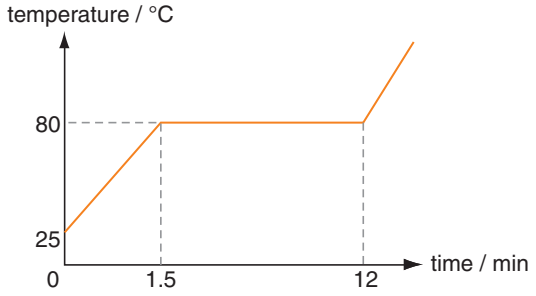


Fig e

(a) Find the specific heat capacity of the solid.

(b) Find the specific latent heat of fusion of the solid.

(c) Does the liquid state of the substance have a specific heat capacity higher than, equal to or lower than that of the solid state? Explain your answer.

*13 The garment steamer shown in Figure f uses steam at 100 °C to remove wrinkles from clothes. Its water tank can fill up to a maximum of 2 kg of water.



Fig f

(a) The water in the water tank is at 20 °C.

(i) Find the total energy required to change all the water in the tank into steam at 100 °C.

(ii) When the garment steamer works, it heats the water with a constant power of 1600 W. Estimate the time required to use up all the water in the tank.

(b) Describe the change in molecular PE and average molecular KE of the water during vaporization.

*14 In an experiment, Steve pours 0.3 kg of water at 20 °C into a glass containing 0.02 kg of melting ice (Fig g). After a while, the mixture reaches thermal equilibrium at 14 °C.



Fig g

(a) Calculate the specific latent heat of fusion l_f of ice using the above data. Given that the specific heat capacity of water is 4200 J kg^{-1} °C^{-1}.

(b) The standard value of l_f is 3.34×10^5 J kg^{-1}.

(i) Compare the result in (a) with the standard value.

(ii) State one possible source of error that causes the difference in (b)(i). Explain your answer.

(iii) Suggest one precaution that can help reduce this error.

4.2 Evaporation and condensation

1 Evaporation and boiling

DSE goal
Realize the occurrence of evaporation below boiling point.

DSE exam
13(1A)Q1

Puddles disappear and wet clothes dry out when water (liquid) changes to vapour (gas) (Fig 4.2a). In both cases, the water temperature is below the boiling point.



Fig 4.2a *(i) A puddle and (ii) wet clothes.*

In fact, a liquid can vaporize through two processes: boiling and **evaporation**. Table 4.2a compares evaporation and boiling.

	Evaporation	Boiling
Differences	It occurs at any temperature below the boiling point.	It occurs at the boiling point.
	It occurs on the liquid surface.	It occurs throughout the liquid.
	No bubbles are formed.	Bubbles are formed inside the liquid.
	Energy is taken from the liquid itself or its surroundings.	Energy is taken from the heat source.
Similarity	**Latent heat of vaporization** is absorbed in both processes.	

Table 4.2a *Comparison between evaporation and boiling.*

2 Cooling effect of evaporation

DSE goal
Explain the cooling effect of evaporation.

Your hands feel cool when you rub them with alcohol-based hand sanitizing gel (Fig 4.2b); you may also feel cold when your body is wet. This is because the liquids take energy away from the skin when they evaporate. This produces a cooling effect. The energy taken away is the latent heat of vaporization.



Fig 4.2b *Experiencing the cooling effect of evaporation.*

evaporation 蒸發

You may observe the cooling effect in the following activity.

Video

→ This video demonstrates the Hands-on activity.

Hands-on activity

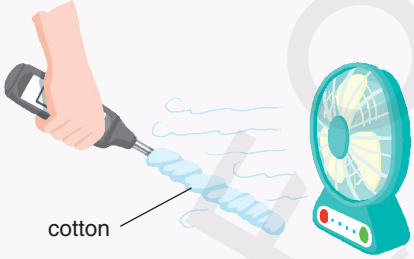
Cooling effect of evaporation

→ Worksheet is available on OUP web.

What you need



What to do



1 Wrap a thermometer with alcohol-soaked cotton.
2 Blow the thermometer with a fan. Observe the temperature drop in 2 minutes.
3 Repeat the steps with dry cotton.

What you find

1 What happens to the alcohol in the cotton after it is blown by the fan for 2 minutes?
2 In which case does the thermometer have a drop in temperature?

Alcohol is flammable. Keep it away from flames and sparks.

DSE goal

Explain evaporation in terms of molecular motion.

DSE exam

12(1A)Q2, 17(1A)Q3

Simulation

→ This simulation shows the motion of molecules near the surface of water to demonstrate evaporation. It is available on OUP web in PC version (.exe) only.

3 Evaporation and molecular motion

a Molecular motion during evaporation

In a liquid, the molecules keep bumping into each other. As a result, some molecules gain KE while other lose KE (Fig 4.2c).

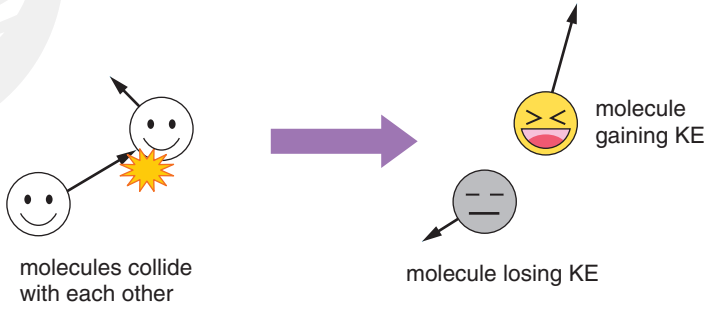


Fig 4.2c *The change in KE among molecules during collision.*

When the **molecules at the surface of a liquid** gain enough KE, they may escape into the space above the liquid and become vapour molecules (Fig 4.2d). In other words, evaporation occurs.

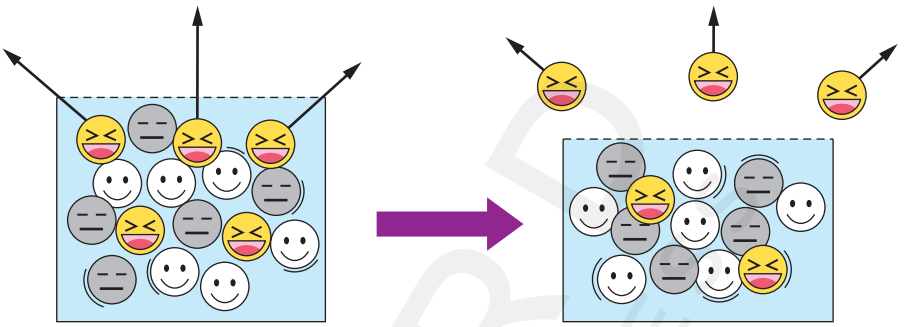


Fig 4.2d Escape of more energetic molecules from the liquid surface.

After molecules with high KE have left the liquid, the remaining molecules have a lower average KE, so the temperature of the liquid drops, causing the cooling effect.

While fast molecules at the liquid surface escape, slow vapour molecules return to the liquid surface. The net rate of evaporation is the difference between the rate of escape and the rate of return.

b Factors affecting evaporation

DSE goal

Discuss the factors affecting rate of evaporation.

Video

→ This video demonstrates Expt 4f.

→ This is a Video Quiz about this experiment.

Experiment 4f / Factors affecting evaporation

1 Wet a cloth and measure its mass.

2 Place the cloth under the hot air blown out of a hairdryer. Find the difference in mass after 5 minutes.



3 Repeat steps 1 and 2 by

Factor: temperature ▶ (a) placing the cloth under cool wind,

Factor: surface area ▶ (b) folding the cloth,

Factor: density of vapour ▶ (c) placing the cloth in a windless environment.

Make sure the wet cloth has the same initial mass.

Discussion

How does each factor affect the rate of evaporation?

The experiment results are summarized in Table 4.2b. A brief explanation describing the effect of each factor is also given.

Factor	Effect on evaporation rate	Explanation
Temperature	Temperature ↑ ⇒ evaporation rate ↑	Under a higher temperature, molecules move faster on average and so more molecules can escape.
Surface area	Surface area ↑ ⇒ evaporation rate ↑	A larger surface area allows more molecules to escape.
Density of vapour	Density of vapour ↑ ⇒ evaporation rate ↓	With a higher density of vapour, vapour molecules have a greater chance of returning to the liquid.

It is the density of vapour (rather than the presence of air) that matters. Evaporation also occurs in a vacuum. ▶

Table 4.2b *Factors affecting evaporation.*

Wind can carry away excess vapour and reduce the density of vapour near the surface of the liquid. Therefore, the rate of evaporation increases in a windy environment (Fig 4.2e).

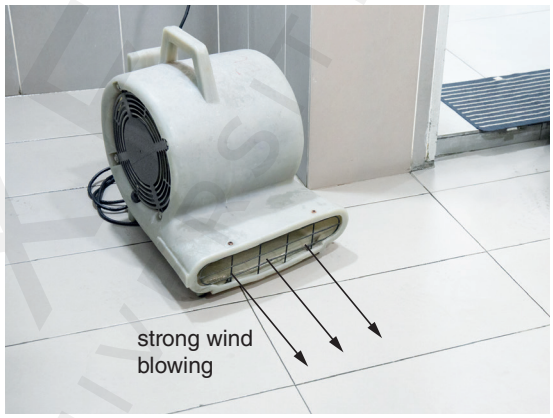


Fig 4.2e *A machine blowing strong wind is used in a washroom to keep the floor dry.*

Example 9 | Cooling effect of sweating

Yvonne of mass 50 kg is exercising and is sweating heavily.

(a) How much energy is taken away by 650 g of sweat when it evaporates? Assume the specific latent heat of vaporization of sweat is 2.26×10^6 J kg^{-1}.

(b) If this amount of energy was not removed from the body, by how much would Yvonne's body temperature increase? Take the average specific heat capacity of the human body to be 3500 J kg^{-1} $°C^{-1}$.

(c) Explain why evaporation occurs in terms of molecular motion.

(d) *Heat stroke* occurs when our body temperature becomes too high. Explain why the risk of getting heat stroke on a humid day is higher.

heat stroke 中暑

Know more

Heat stroke
Do you know what should be done if someone has heat stroke?
▸ p.135

Relative humidity
Relative humidity is a measure of water vapour density in air. Do you know how to determine this value?
▸ p.135

Solution

(a) Energy taken away = ml_v
$= 0.65 \times (2.26 \times 10^6)$
$= 1.469 \times 10^6 \approx 1.47 \times 10^6$ J

(b) By $Q = mc\Delta T$,
$1.469 \times 10^6 = 50 \times 3500 \times \Delta T$
$\Delta T = 8.39$ °C
Yvonne's body temperature would increase by 8.39 °C.

(c) Some molecules at the surface of a liquid gain enough KE. They escape into the space above the liquid and become vapour molecules.

(d) Sweat evaporates slowly on a humid day. As such, energy cannot be removed from a human body through the evaporation of sweat effectively. Therefore, the risk of getting heat stroke is higher.

4 Condensation

If you are wearing glasses while drinking hot drink, your glasses may fog up (Fig 4.2f). This is because the water vapour from the hot drink is cooled suddenly by the glass surface and turns into water droplets due to condensation (Fig 4.2g).



Fig 4.2f *Foggy glasses.*

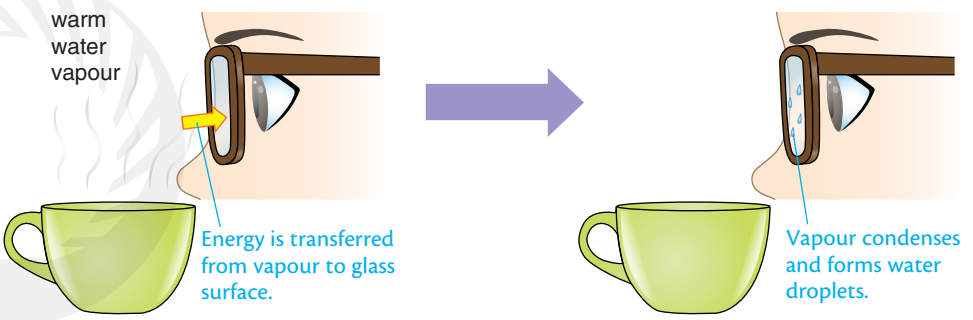


Fig 4.2g *Warm water vapour condenses on the cold glass surface to make it foggy.*

The glass surface is cooler than the water vapour from the hot drink in above case. ▶

Condensation can occur at or below boiling point. It occurs readily when vapour meets a cooler surface. During this process, latent heat of vaporization is released by the vapour.

relative humidity 相對濕度

STSE

Heating and cooling using latent heat

There are some cooling and heating systems that make use of latent heat.

→ This simulation demonstrates the working principle of the ice storage cooling system. It is available on OUP web in PC version (.exe) only. **Simulation**

Ice storage cooling systems

An ice storage cooling system (Fig a) produces cool air by making ice at night. When the ice melts during the daytime, air is cooled and piped through a building. This can save electricity and reduce pollution.

The power demand is much higher in daytime than at night. If a cooling system works in daytime, power plants have to increase the power output. A system working at night will not increase the power demand in daytime and so the power plants can work with a steady power output. This saves electricity and reduces pollution.

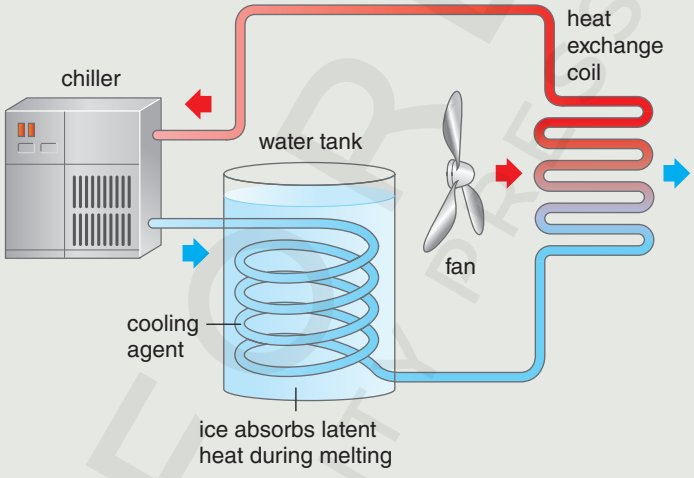


Fig a

Latent heat recovery systems

A latent heat recovery system (Fig b) works in the opposite way to an ice storage cooling system. During the daytime, the solar energy is used to melt a wax-based substance. At night, the substance freezes and releases latent heat to heat up water/space.

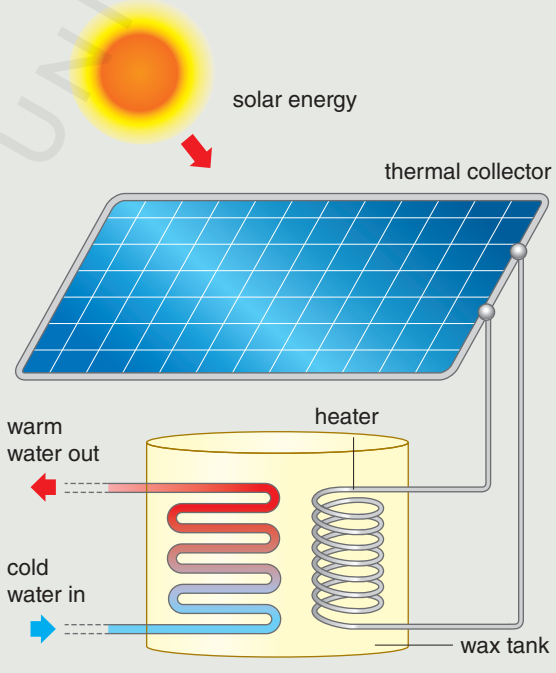


Fig b

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 A bowl of soup with a lid (Fig a) takes a (longer / shorter) time to cool down because the lid (enhances / slows down) the evaporation of the soup.



Fig a

2 Which of the following statements correctly state(s) the difference between boiling and evaporation? Put a tick in the corresponding box(es).

- ☐ Latent heat of vaporization is taken away from a liquid during boiling but not during evaporation.
- ☐ Boiling always occurs at a definite temperature but evaporation does not.
- ☐ Boiling takes place on the surface, but evaporation occurs throughout the liquid.
- ☐ Bubbles are formed violently during boiling, but slowly during evaporation.

3 Which of the following statements about the evaporation of a liquid must be correct? Put a tick in the corresponding box(es).

- ☐ During evaporation, all molecules at the liquid surface gain enough potential energy to escape into the space above.
- ☐ The average KE of the molecules in the remaining liquid decreases during evaporation.
- ☐ A liquid does not evaporate on a windless day.

4 Figure b shows a cup of hot tea.



Fig b

Which of the following actions may increase the rate of evaporation of the tea? Put a tick in the corresponding box(es).

- ☐ Add some ice to the tea.
- ☐ Pour the tea into another cup with a wider diameter.
- ☐ Use a fan to blow the cup of tea.

5 *True or false:* When water vapour condenses on a surface, the surface is warmed. (T/F)

Exercise 4.2

Take c (water) = 4200 J kg^{-1} °C^{-1}

l_f (ice) = 3.34×10^5 J kg^{-1}

l_v (water) = 2.26×10^6 J kg^{-1}

1 Which of the following would make wet clothes dry more slowly?

A Folding the clothes

B Low humidity

C Wind

D Exposure to sunlight

***2** Which of the following best explains why evaporation causes a liquid to cool?

A When a change of state takes place, the PE of the liquid molecules decreases.

B The amount of liquid decreases during evaporation, resulting in a decrease in the total KE of the liquid molecules.

C Faster molecules in the liquid escape during evaporation, resulting in a lower average KE of the remaining liquid molecules.

D The liquid molecules at the liquid surface lose KE while colliding with air molecules, which results in a decrease in the average KE of the liquid molecules.

*3 Which of the following statements about vaporization is correct?

A Boiling is a kind of vaporization.

B The vaporization of water occurs only at 100 °C.

C Bubbles must be formed in vaporization.

D Vaporization takes place on the surface of the substance only.

4 On a hot and humid day, Kelvin switches on the air conditioner and keeps the room temperature at 25 °C. After some time, a total of 10 g of water vapour condenses on the outer surface of the glass window (Fig a).



Fig a

(a) Find the energy released when 10 g of water vapour condenses on the glass window.

(b) Explain how the internal energy of the water vapour changes when it condenses.

*5 Carman is getting out of a swimming pool (Fig b).



Fig b

(a) Explain why she feels cold when getting out.

(b) Explain why she would feel even colder if there is a strong wind.

*6 (a) It takes a longer time to heat up a pot of water when the pot is not covered (Fig c). Explain this by considering the effect of evaporation.



Fig c

(b) White fog can be seen above the water during heating. What is the white fog? How is it formed?

*7 After taking a swim, the water on Sam's skin evaporates.

(a) Explain why evaporation occurs in terms of molecular motion.

(b) If 50 g of water on his skin evaporates, how much energy is taken away?

(c) Suppose this amount of energy is taken from Sam's body only. Ignoring the effect of other temperature regulation systems, by how much would his body temperature decrease? The mass of Sam is 65 kg and the average specific heat capacity of a human body is 3500 J kg^{-1} $°C^{-1}$.

(d) Give two factors that affect the rate of evaporation.

*8 Two identical glasses *A* and *B* contain the same amount of hot water at the same temperature. Glass *A* is half covered by a plastic sheet while glass *B* is uncovered (Fig d). After several hours, the water level in glass *A* is higher than that in glass *B*. Explain the reason.

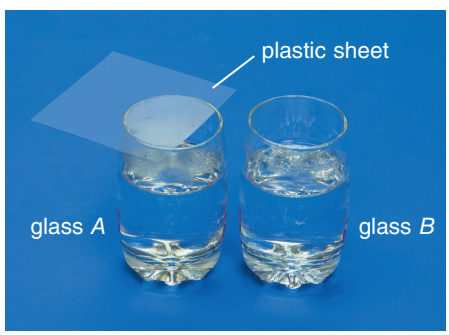
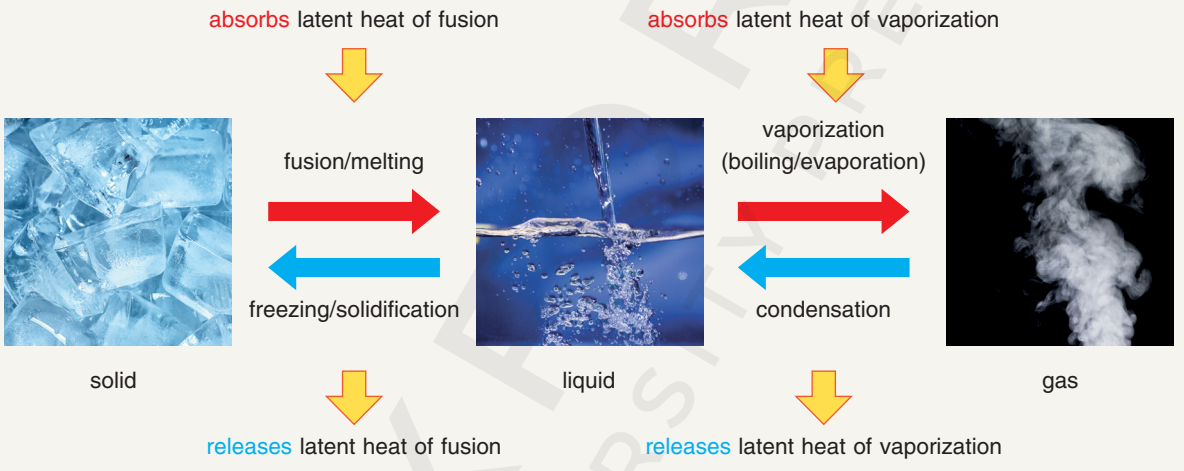


Fig d

Key concepts 4

1 Change of state and latent heat

- When a substance changes its state at the boiling point or melting point,
  - latent heat is absorbed or released;
  - temperature remains unchanged.
- Evaporation and condensation can occur below the boiling point.



2 Specific latent heat *l*

- The energy absorbed or released to change the state of 1 kg of the substance without a change in temperature

$$l = \frac{Q}{m}$$

- Unit: J kg^{-1}
- l_f: specific latent heat of fusion
- l_v: specific latent heat of vaporization

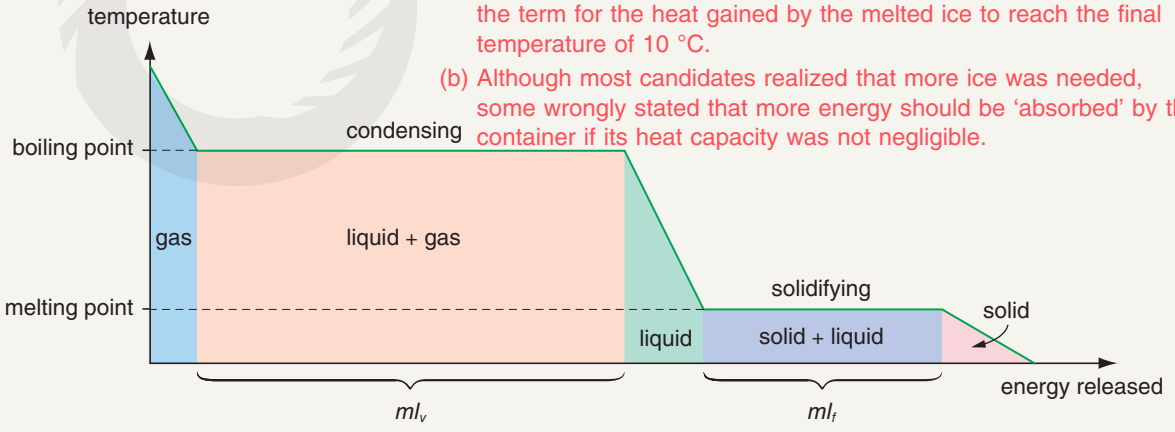
Q27 Exam Report: Parts (a) and (c) were well answered but only the more able candidates managed to answer part (b). In (a), some candidates wrongly used 4200 J kg^{-1} °C^{-1} for the specific heat capacity of steam. In (b), many candidates failed to take into account the energy released by water from 100 °C to temperature *T* in their calculation.

Q28 Exam Report:

(a) Most candidates obtained the correct answer though a few omitted the term for the heat gained by the melted ice to reach the final temperature of 10 °C.

(b) Although most candidates realized that more ice was needed, some wrongly stated that more energy should be 'absorbed' by the container if its heat capacity was not negligible.

Cooling curve of a substance



3 Internal energy

Internal energy = total molecular KE + total molecular PE

= no. of molecules × (average molecular KE + average molecular PE)

Factors affecting internal energy

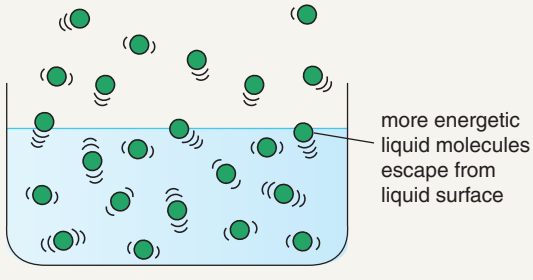
	Average molecular KE	Average molecular PE	Internal energy
Temperature ↑	↑	unchanged	↑
Change of state (solid ⇒ liquid ⇒ gas)	unchanged	↑	↑
Mass ↑	unchanged	unchanged	↑

4 Evaporation

- Change of state from liquid to gas at temperatures below boiling point
- Evaporation ⇒ latent heat is absorbed from surroundings ⇒ cooling effect

Molecular motion during evaporation

- Molecules in the liquid collide with each other
  ⇒ Molecules at liquid surface gains enough KE
  ⇒ Escape into the space above the liquid
  ⇒ Become vapour molecules
  ⇒ Evaporation occurs
- Molecules with less KE stay in the liquid
  ⇒ Average KE of molecules ↓
  ⇒ Temperature of liquid ↓



Factors affecting rate of evaporation

- Temperature ↑ ⇒ rate ↑
- Surface area ↑ ⇒ rate ↑
- Density of vapour ↑ ⇒ rate ↓

Condensation

- Change of state from gas to liquid at or below boiling point
- Latent heat is released

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Key terms

1 boiling 沸騰 p.93
2 condensation 凝結 p.97
3 cooling curve 冷卻曲線 p.96
4 evaporation 蒸發 p.118
5 freezing/solidification 凝固 p.96
6 fusion 熔解 p.93
7 heating curve 加熱曲線 p.94
8 latent heat 潛熱 p.93
9 potential energy 勢能 p.113
10 specific latent heat 比潛熱 p.98
11 specific latent heat of fusion 熔解比潛熱 p.99
12 specific latent heat of vaporization 汽化比潛熱 p.99
13 vaporization 汽化 p.94

Revision 4

Take c (water) = 4200 J kg^{-1} °C^{-1}
l_f (ice) = 3.34×10^5 J kg^{-1}
l_v (water) = 2.26×10^6 J kg^{-1}

Multiple-choice questions

★1 4.2 A glass of water at room temperature evaporates. Which of the following would reduce the rate of evaporation?

(1) Cover the glass of water with a lid.
(2) Place the water in the sun.
(3) Make the room cooler.

A (1) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

★2 4.2 Which of the following statements about evaporation is/are correct?

(1) A liquid may absorb latent heat from its surroundings during evaporation.
(2) The temperature of the remaining liquid always increases after evaporation.
(3) Water evaporates faster in a high-humidity environment.

A (1) only
B (1) and (2) only
C (1) and (3) only
D (2) and (3) only

★3 4.1 A substance is heated at a constant rate and the variation of its temperature with time is shown in Figure a.

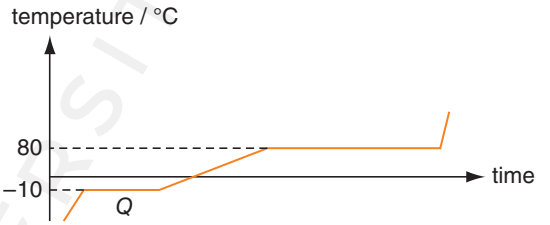


Fig a

Which of the following statements about this substance is correct?

A Its specific latent heat of vaporization is smaller than its specific latent heat of fusion.
B Its specific heat capacity is larger when it is in a solid state than in a gaseous state.
C Its melting point is 80 °C.
D No solid exists in stage Q.

★4 4.1 A substance changes from solid to liquid when it is heated at a constant rate. The following graphs show how different kinds of energy in the substance vary with time (Fig b).

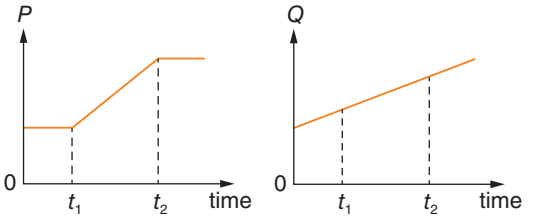


Fig b

What are P and Q?

	P	*Q*
A	Average molecular KE	Average molecular PE
B	Average molecular PE	Average molecular KE
C	Average molecular KE	Internal energy
D	Average molecular PE	Internal energy

★5 4.1 The graph below shows the energy supplied to vaporize a certain amount of liquid X at its boiling point (Fig c).

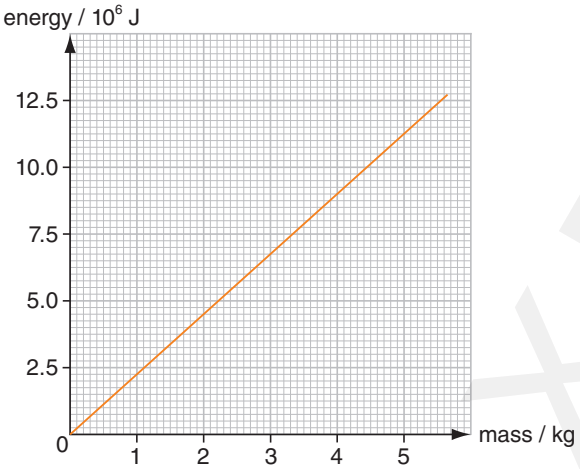


Fig c

Which of the following quantities can be found from the graph?

(1) Specific heat capacity of X

(2) Specific latent heat of vaporization of X

(3) Boiling point of X

A (2) only B (1) and (2) only

C (1) and (3) only D (2) and (3) only

★6 4.1 An immersion heater is used to heat a substance of mass m. Figure d shows how the temperature of the substance varies with time.

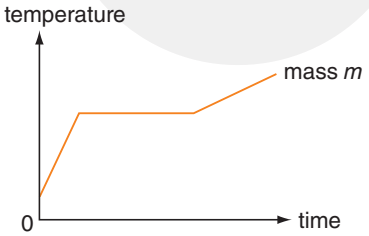
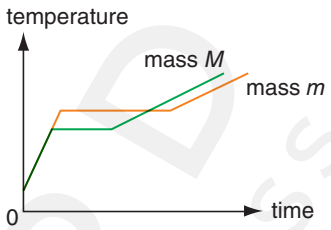


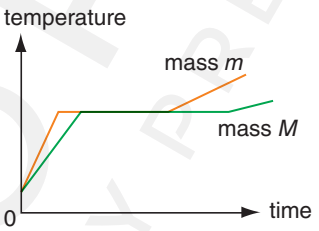
Fig d

The immersion heater is then used to heat a larger mass M of the same substance. Which of the following graphs (green lines) best represents the variation in temperature in this case?

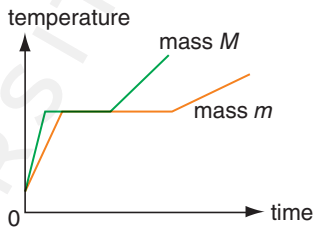
A



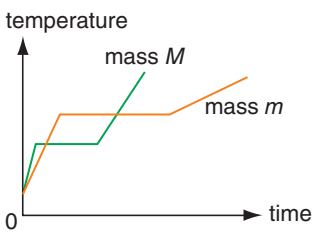
B



C



D



★7 4.1 A beaker of water is heated at a constant power. After a period of time, the water starts to boil. Suppose the steam escapes from the beaker completely without condensing. The variation of mass m of the water in the beaker against time t is shown in Figure e.

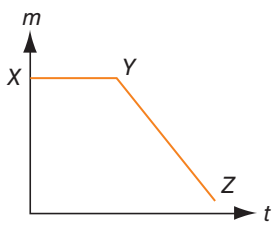


Fig e

If the power of the heater increases, how would the graph change?

	Segment *XY*	Segment *YZ*
A	shorter	steeper
B	longer	steeper
C	shorter	flatter
D	longer	flatter

★★8 4.1 An ice cube at 0 °C is mixed with steam at 100 °C. After a while, they reach thermal equilibrium and become water at 50 °C. What is the ratio of the mass of ice to the mass of steam at the beginning?

A 1 : 1 B 3 : 1

C 4.5 : 1 D 6.8 : 1

Refer p.107

4.1 **9 HKDSE 2014 Paper 1A Q2**

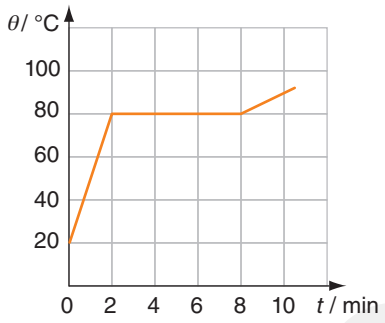


Fig f

An electric heater of constant power is used to heat a solid substance *X* which is insulated from the surroundings. The variation of its temperature θ with time t is shown above. *X* has a specific heat capacity of 800 J kg^{-1} $°C^{-1}$ in its solid state. What is the specific latent heat of fusion of *X*?

A 144 kJ kg^{-1} (55%)

B 192 kJ kg^{-1}

C 202 kJ kg^{-1}

D Answer cannot be found as both the mass of *X* and the power of the heater are not known.

4.1 **10 HKDSE 2016 Paper 1A Q2**

0.3 kg of water at temperature 50 °C is mixed with 0.2 kg of ice at temperature 0 °C in an insulated container of negligible heat capacity. What is the final temperature of the mixture?

A −1.8 °C (26%) B 0 °C (39%)

C 1.8 °C (22%) D 3.0 °C (13%)

Q10 Exam Report: About one quarter of the candidates chose option A which is physically not possible.

Misc **11 HKDSE 2018 Paper 1A Q2**

In some countries, the outdoor temperature can drop below 0 °C in winter such that the lake surface becomes a thicker layer of ice. However the water beneath the ice surface does not as easily to turn into ice and most aquatic life can survive during winter time.

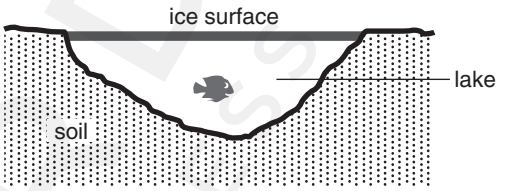


Fig g

Which statement below best explains this phenomenon?

A The layer of ice provides good heat insulation. (55%)

B The freezing point of water below the ice surface is much lower than 0 °C.

C There is thermal energy transferred from the soil to the water in the lake.

D Ice releases latent heat when it melts.

4.1 **12 HKDSE 2021 Paper 1A Q1**

Solid substances *X* and *Y* of equal mass are heated by heaters of power 2*P* and *P* respectively. The graph shows how the temperature θ of each substance varies with the heating time *t*.

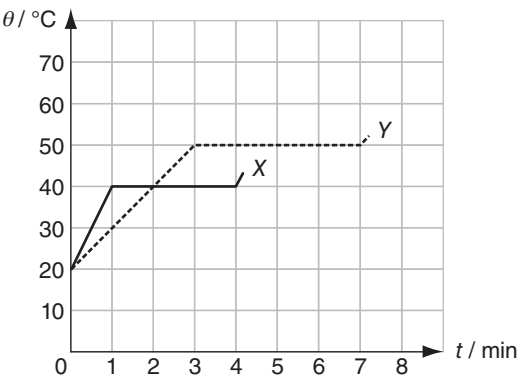


Fig h

What is the ratio of the specific latent heat of fusion of *X* to that of *Y*?

A 3 : 2 (62%)

B 3 : 4

C 4 : 3

D 2 : 3

Conventional questions

★13 4.1 (a) Water of mass 0.2 kg at 10 °C is put into a freezer to make ice at 0 °C. How much energy is released by the water? (2 marks)

(b) Farmers in cold countries spray fruit trees (Fig i) with water to prevent the inside of the fruit from freezing. Explain how this method works. (3 marks)



Fig i

★14 4.1 Some ice cubes are left at room temperature (Fig j). The graph below shows how the temperature of the ice changes with time t (Fig k).



Fig j

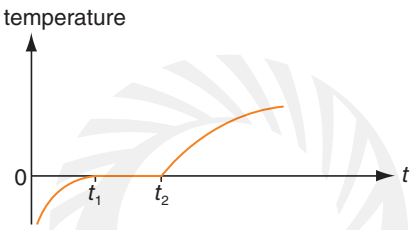


Fig k

(a) Describe how the average PE and the average KE of the ice molecules change

(i) from $t = 0$ to t_1, and

(ii) from $t = t_1$ to t_2.

(2 marks)

(b) Tom claims that the internal energy of the ice cubes remains constant from $t = t_1$ to t_2 since the temperature is unchanged. Comment on this statement. (2 marks)

★15 4.2 A dog cools itself by panting (Fig l). Energy is taken away from it when water on its tongue evaporates.



Fig l

(a) Suppose 10 g of water evaporates from the tongue in 1 minute and all the latent heat comes from the dog. Estimate the average rate at which energy is taken away from the dog. (2 marks)

(b) Suggest one action that the dog can take to increase the evaporation rate of water on its tongue. (1 mark)

★16 4.2 A glass of cold water is left at room temperature (Fig m). A total of 3 g of water vapour condenses on the surface of the glass.



Fig m

(a) Find the energy released when 3 g of water vapour condenses. (2 marks)

(b) What is the increase in temperature of the glass of water due to the condensation of the water vapour? Take the heat capacity of the glass together with the water inside to be 2000 J $°C^{-1}$. (2 marks)

(c) When the same glass of cold water is put on an outdoor table, water vapour does not condense on the glass surface. Suggest a possible reason and explain your answer. (2 marks)

★17 Misc (a) A beaker is filled with 50 g of water at 25 °C. How much energy is needed to heat up and vaporize all the water? (2 marks)

(b) The water in the beaker is accidentally poured onto the floor. After a few days, the water disappears. Assume the temperature of the surroundings is also 25 °C.

(i) Name the phenomenon that occurs. (1 mark)

(ii) Explain, in terms of molecular motion, why this phenomenon occurs. (2 marks)

(iii) Estimate the total energy absorbed by the water during the process. (2 marks)

★18 4.2 On a very hot day, Kenny sweats 0.5 kg of water in 1 hour (Fig n). Assume that all the sweat evaporates and all latent heat is taken from his body.



Fig n

(a) Explain why evaporation occurs in terms of molecular motion. (2 marks)

(b) Find the amount of energy removed from the body by the sweat in 1 hour. (2 marks)

(c) At what rate is the energy removed from the body through sweating? (2 marks)

(d) Give two factors that affect the rate of evaporation. (2 marks)

★19 4.1 A refrigerator converts 1.5 kg of water at 30 °C into ice at –5 °C in 5 hours.

Given: the specific heat capacity of ice is 2060 J kg^{-1} $°C^{-1}$

(a) Find the energy released by the water during the process. (2 marks)

(b) Assume that energy is taken away from the water at a constant rate.

(i) Find this rate in watts. (2 marks)

(ii) Sketch the temperature–time graph of the water. (3 marks)

★20 4.2 A pot of hot water is covered with a lid (Fig o).



Fig o

(a) Explain briefly why water droplets form on the inner surface of the lid. (2 marks)

(b) Suppose the process in (a) will cause the temperature of the lid to increase by 35 °C. Find the mass of the water droplets forming in the process in (a). Take the heat capacity of the lid to be 520 J $°C^{-1}$. (2 marks)

★21 4.1 Maria heats some crushed solid of mass 0.8 kg using a heater with a power of 350 W.

Given: melting point of the solid = 90 °C

specific heat capacity of the solid = 1300 J kg^{-1} $°C^{-1}$

specific latent heat of fusion of the solid = 6.10×10^4 J kg^{-1}

initial temperature of the solid = 25 °C

(a) The heater is turned on at time $t = 0$.

(i) When does the solid start to melt? (2 marks)

(ii) When does the solid melt completely? (2 marks)

(b) Using the result in (a), sketch the heating curve of the solid from $t = 0$ to some time after the solid has completely melted. In your diagram, indicate

(i) the melting point,

(ii) the time when the solid starts to melt,

(iii) the time when the solid melts completely. (2 marks)

(c) In the graph in (b), sketch a new heating curve if the amount of solid used was halved. (2 marks)

(d) Is the actual time that the solid completely melts earlier or later than that calculated in (a)? Explain. (2 marks)

★22 4.2 Read the following passage about an **evaporative cooler** and answer the questions that follow.

By making use of the cooling effect of evaporation, we can cool the air without using an air conditioner. Figure p shows a simple evaporative cooler. The cooler takes in hot and dry air from the outside using a fan. As the air passes through a pad, which is kept wet by a pump, it becomes cooler since some of the energy of the hot air is used to evaporate the water on the pad. The cool air is then given out to the room.

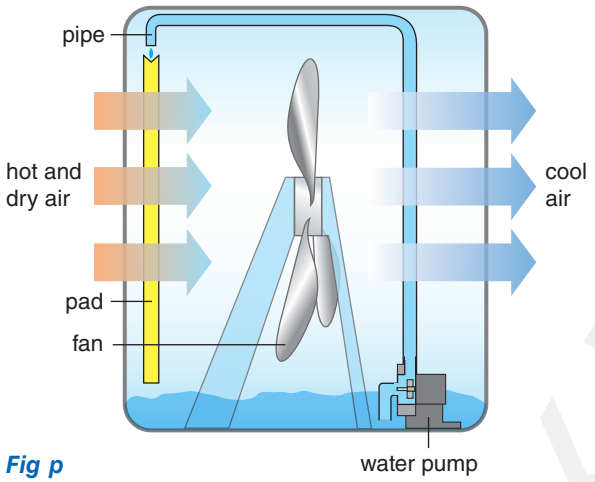


Fig p

(a) State one required condition about the air outside the room for this evaporative cooler to work effectively. Explain your answer. (2 marks)

(b) State and explain one drawback of this evaporative cooler. (2 marks)

(c) Is it possible to cool the air using a fan only? Explain your answer. (2 marks)

★23 4.1 Susan uses an immersion heater to heat a beaker of water on a balance (Fig q). Figure r shows how the balance reading changes as the water is heated. You may ignore the energy loss to the surroundings and the effect of evaporation.

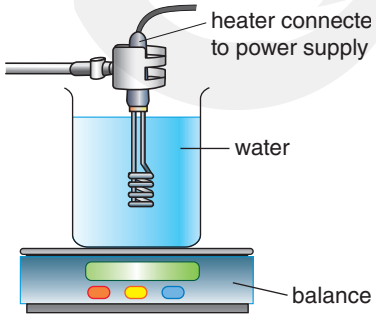


Fig q

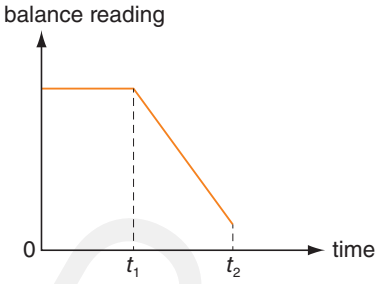


Fig r

(a) Sketch a graph to show how the temperature of water in the beaker varies with time. Briefly explain your answer. (3 marks)

(b) In Figure r, sketch a new curve for the case if the beaker contained more water. Briefly explain your answer. (4 marks)

★24 4.1 A certain liquid is heated by an immersion heater that supplies energy at a rate of 20 J s^{-1} (Fig s). The liquid is brought to its boiling point and kept boiling for 3 minutes. The mass of the liquid reduces by 10 g in these 3 minutes.



Fig s

(a) Calculate the specific latent heat of vaporization l_v of the liquid. (2 marks)

(b) (i) The standard value of l_v of the liquid is 3.00×10^5 J kg^{-1}. If the difference between the value in (a) and the standard value is solely due to the energy loss to the surroundings, find the energy lost to the surroundings. (2 marks)

(ii) Suggest a precaution to reduce the energy loss to the surroundings. (1 mark)

(c) Suggest a possible source of error other than the energy loss to the surroundings. Explain how this error affects the value of l_v obtained. (3 marks)

**25 4.1 Water of 250 g is heated by an immersion heater with a constant power of 300 W. Initially, the water is at 50 °C. At time t_1, the water starts to boil. At time t_2, half of the water has vaporized.

(a) Find the value of t_1 and t_2. (4 marks)

(b) Sketch the following graphs. Mark the time instants t_1 and t_2 in the graphs.

(i) Temperature–time graph

(ii) Mass–time graph

(4 marks)

Refer p.94–95

**26 4.1 You are given the following apparatus and materials:

- A polystyrene cup
- Crushed melting ice
- An immersion heater of power P
- A stop-watch
- An electronic balance
- A power supply unit

Assume the polystyrene cup is highly insulating.

(a) Describe how you should use the given apparatus and materials to estimate the specific latent heat of fusion l_f of ice. Your description should include the physical quantities to be measured and how you would obtain the result from the measurements. (4 marks)

(b) Give a possible source of error. Explain how this error will affect the value of l_f obtained. (3 marks)

Refer p.101–102

27 4.1 **HKDSE 2012 Paper 1B Q1** See p.126 for the Exam Report.

Cappuccino is an Italian style coffee topped with a layer of frothy milk (Fig t).



Fig t

Frothy milk is made by bubbling steam through milk, which is held in a metallic jug (Fig u). Steam is ejected from the steam wand of a cappuccino machine (Fig v).



Fig u *Fig v*

Given:

specific latent heat of vaporization of water = 2.26×10^6 J kg^{-1}

specific heat capacity of water = 4200 J kg–1 °C–1

specific heat capacity of steam = 2000 J kg–1 °C–1

specific heat capacity of milk = 3900 J kg^{-1} $°C^{-1}$

(a) Calculate the total amount of heat released when 20 g of steam at 110 °C cools to 100 °C and condenses to water at 100 °C. (3 marks)

(b) 20 g of steam at 110 °C is bubbled through 200 g of milk at 15 °C to make frothy milk. Using the result in (a), estimate the temperature of the frothy milk. (2 marks)

(c) Would the actual temperature of frothy milk be higher than, equal to or lower than the value found in (b)? Explain. (2 marks)

28 4.1 **HKDSE 2019 Paper 1B Q1(a)** See p.126 for the Exam Report.

An insulated container of negligible heat capacity contains 1.5 kg of tea at a temperature of 60 °C.

(a) What mass of ice at 0 °C should be added to the tea so that the final temperature of the mixture is lowered to 10 °C? Assume that the specific heat capacity of tea is the same as that of water. (3 marks)

(b) If the heat capacity of the container is not negligible, explain whether more ice, less ice or the same amount of ice is needed to obtain the final temperature of 10 °C. (2 marks)

Know more

Heat stroke (p.122)

Heat stroke is likely to occur if one's body temperature gets too high, especially when engaging in outdoor activities on a hot summer day.

People suffering from heat stroke cannot sweat normally. Consequently, their body temperature rises (possibly higher than 40.6 °C) and their skin becomes flushed, hot and dry. In more serious cases, unconsciousness or even death may result. People suffering from heat stroke must immediately be cooled. This can be done by moving them to a cool place, sponging them with cool water, and fanning them vigorously (Fig a). Fanning helps increase the rate of evaporation to take away the excess energy from their body.

Even if the air temperature is the same, you will feel hotter on a humid day than on a dry one. Therefore, scientists have introduced the heat stress index, which combines the effect of air temperature, relative humidity and other factors, to determine the 'apparent temperature' (Fig b).



Fig a

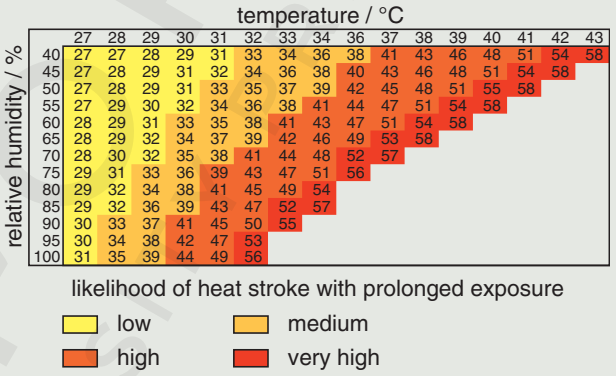


Fig b

Relative humidity (p.122)

Relative humidity is the amount of water vapor actually in the air, expressed as a percentage of the maximum amount of water vapor the air can hold at the same temperature. It affects the rate of evaporation and hence the cooling effect of evaporation. Therefore, by finding out how effective the cooling effect is, we can determine the relative humidity.

The set-up (Fig c) for measuring relative humidity consists of two identical thermometers, a dry one *D* and a wet one *W*. The bulb of *W* is covered by a wet cloth, which is partially immersed in water. Due to the cooling effect, *W* would measure a lower temperature than *D*. A lower relative humidity results in a greater temperature difference between the two thermometers. The relative humidity can be determined from the readings of both thermometers.

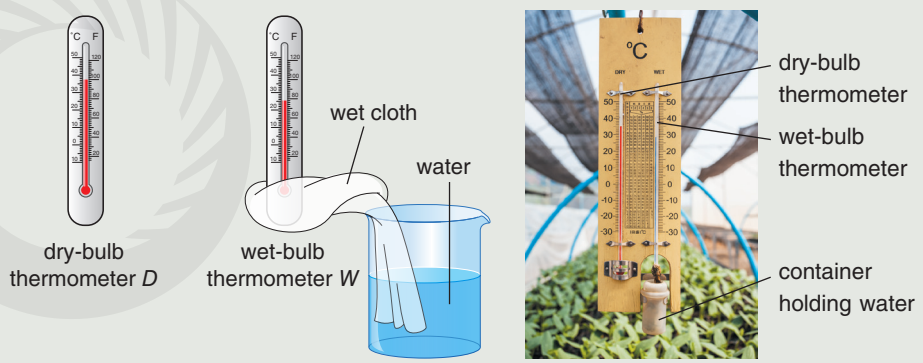


Fig c

Physics in cooking

Simmering



Simmering is a common method used to make soup and desserts. First, water and the ingredients are put in a container with a lid. The container is in turn placed in a large pot of boiling water over the stove. You may try the activity below to learn about the principle of simmering.

Hands-on activity

→ Worksheet is available on OUP web.

Water that does not boil!

What you need



What to do



1 Put the ceramic mug into the cooking pot. Half-fill them both with water.

2 Heat the pot on a cooking stove. Observe whether the water boils.

The set-up is very hot during heating. Do not touch it.

What you find

Does the water boil both inside and outside the mug?

→ This video demonstrates the Hands-on activity.

Video

In the activity above, the water inside the ceramic mug is heated by the water outside. Both the water inside and outside the mug eventually reach 100 °C, the boiling point of water. As their temperatures are the same, there is no more energy transfer and hence no latent heat is supplied for the water inside the mug to change into steam. Therefore, boiling does not occur in the mug.

When simmering soup, since the water in the container never boils, the flavor and the essence of the ingredients are retained in the soup.



Steaming

Steaming is a method of cooking using steam. It is typically used in Chinese cuisine. This can be done with an electric steamer or a wok containing boiling water. Steam releases a large amount of latent heat when it condenses on the food. This cooks the food quickly and evenly.